CLINICAL CONDITIONS

Gastrointestinal Practice Tests

3 tests of 50 USMLE-style questions, with answers and explanations

For educational use only. This is a study aid, not a clinical reference.

Instructions

60 minutes
for each test

50 questions
in each test

About 72 seconds
for each question

Taking a test

- Take each test in one sitting, with a timer set to 60 minutes. Do not pause the timer.
- Each question has five options and one best answer. Choose the single best option, even if more than one seems reasonable.
- Mark your answers on the answer sheet after the last question of each test, or on a separate piece of paper.
- Do not use notes, the website, or other references. The abbreviation list on the next page is allowed.
- There is no penalty for guessing, so answer every question. An unanswered question counts as incorrect.
- When time ends, stop, even if you have not finished.

About the tests

- No question appears in more than one test, so the three tests can be taken in any order.
- Every test has 15 physiology and 35 clinical questions, covering all 4 physiology topics and all 30 clinical topics, with a similar mix of Step 1 questions (mechanisms, microbiology, and pathology) and Step 2 questions (diagnosis, next step, and management).
- The tests use 150 of the 387 questions on the website, so the website quiz has more practice.
- Questions are in random order, and the topic being tested is not shown until the answers.

Scoring and review

- Your score is the number of correct answers. Use the answer key at the start of each answer section, and write your score on your answer sheet.
- Then read the explanations. Each one names the topic and step, so you can see what to revisit.
- Answers follow the course material. Where a guideline differs, the explanation says so under "Additional information".

Contents

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Abbreviations

You may refer to this page during a test.

5-HT 5-hydroxytryptamine (serotonin)

5-HT3 Serotonin receptor 3, an ion channel

5-HT4 Serotonin receptor 4

ABX Antibiotics

AFP Alpha-fetoprotein

ALP Alkaline phosphatase

ALT Alanine aminotransferase

AMA Anti-mitochondrial antibody

anti-CCP Anti-cyclic citrullinated peptide antibody

anti-dsDNA Anti-double-stranded DNA antibody

anti-HBc Antibody to hepatitis B core antigen

anti-HDV Antibody to hepatitis D virus

anti-tTG Anti-tissue transglutaminase antibody

AQP4 Aquaporin-4

ASIS Anterior superior iliac spine

AST Aspartate aminotransferase

ATP7A Copper-transporting ATPase A

ATP7B Copper-transporting ATPase B

BUN Blood urea nitrogen

cAMP Cyclic adenosine monophosphate

cAMP Cyclic adenosine monophosphate

CCK Cholecystokinin: released by the duodenum in response to fat; makes the gallbladder contract

CD4 Cluster of differentiation 4

CN Cranial nerve

CN X Cranial nerve X (the vagus nerve)

COPD Chronic obstructive pulmonary disease

CRP C-reactive protein

CSF Cerebrospinal fluid

CT Computed tomography

CT AP Computed tomography of the abdomen and pelvis

CTR1 Copper transporter 1

CTZ Chemoreceptor trigger zone

CYP2E1 Cytochrome P450 2E1

CYP7A1 Cholesterol 7α-hydroxylase

D2 Dopamine receptor 2

DMT1 Divalent metal transporter 1

DVT Deep vein thrombosis

ECL Enterochromaffin-like

ERCP Endoscopic retrograde cholangiopancreatography

FAST Focused assessment with sonography in trauma

FBC Full blood count

FODMAP Fermentable oligosaccharides, disaccharides, monosaccharides, and polyols

GE Gastroesophageal

GERD Gastroesophageal reflux disease

GGT Gamma-glutamyl transferase

GIP Glucose-dependent insulinotropic peptide

GLUT2 Glucose transporter 2

GLUT5 Glucose transporter 5

H1 Histamine receptor 1

H2 Histamine receptor 2

H2R Histamine H2 receptor

HBIG Hepatitis B immune globulin

HBsAg Hepatitis B surface antigen: current infection

HBV Hepatitis B virus

HCl Hydrochloric acid

HCV Hepatitis C virus

HDV Hepatitis D (delta) virus

HELLP Hemolysis, elevated liver enzymes, low platelets

HFE Hereditary hemochromatosis gene

HIDA Hepatobiliary iminodiacetic acid scan (cholescintigraphy)

HIV Human immunodeficiency virus

HIV Human immunodeficiency virus

HLA Human leukocyte antigen

IBD Inflammatory bowel disease

IBS Irritable bowel syndrome

ICU Intensive care unit

IF Intrinsic factor

IgA Immunoglobulin A

IgG Immunoglobulin G

IgG4 Immunoglobulin G4

IgM Immunoglobulin M

IM Intramuscular

IM Intramuscular

INR International normalized ratio

IV Intravenous

LCHAD Long-chain 3-hydroxyacyl-CoA dehydrogenase, an enzyme of fatty acid beta-oxidation

LDH Lactate dehydrogenase

LDL Low-density lipoprotein

LDL Low-density lipoprotein

LES Lower esophageal sphincter

LFTs Liver function tests

LLQ Left lower quadrant

M3 Muscarinic acetylcholine receptor 3

MALT Mucosa-associated lymphoid tissue

MASLD Metabolic dysfunction-associated steatotic liver disease (formerly NAFLD)

MCAD Medium-chain acyl-CoA dehydrogenase

MCV Mean corpuscular volume

MRCP Magnetic resonance cholangiopancreatography

MRI Magnetic resonance imaging

MRP2 Multidrug resistance-associated protein 2

NAC N-acetylcysteine

NAPQI N-acetyl-p-benzoquinone imine

NK1 Neurokinin-1 receptor

NSAID Nonsteroidal anti-inflammatory drug

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NSAIDs Nonsteroidal anti-inflammatory drugs

NTS Nucleus tractus solitarius

p-ANCA Perinuclear anti-neutrophil cytoplasmic antibody

PAS Periodic acid-Schiff

PBC Primary biliary cholangitis (formerly primary biliary cirrhosis)

PCR Polymerase chain reaction

PepT1 Peptide transporter 1

PET Positron emission tomography

PG Prostaglandins

PPI Proton pump inhibitor

PPIs Proton pump inhibitors

PT Prothrombin time

PYY Peptide YY

RNA Ribonucleic acid

SGLT1 Sodium-glucose cotransporter 1

SMA Superior mesenteric artery

TIPS Transjugular intrahepatic portosystemic shunt

TPMT Thiopurine methyltransferase

TPN Total parenteral nutrition

UGT UDP-glucuronosyltransferase

VKOR Vitamin K epoxide reductase

WBC White blood cells

Test A

50 questions · 60 minutes

1. A 71-year-old woman has fever of 39.2°C, right upper quadrant pain, and jaundice. Over several hours she becomes confused, and her blood pressure falls to 82/48 mm Hg despite IV fluids. Ultrasound shows a dilated common bile duct with a stone.

Which is the most important definitive step after starting IV ABX and resuscitation?

A. HIDA scan
B. Oral ABX and outpatient follow-up
C. Ursodeoxycholic acid
D. Urgent ERCP for biliary drainage
E. Elective cholecystectomy in 6 weeks

2. A 56-year-old woman with primary biliary cholangitis has fingers that turn white in the cold, tight skin over her fingers, small dilated blood vessels on her face, and difficulty swallowing.

Which antibody is most likely positive for her second condition?

A. Anti-Jo-1 antibody
B. Anti-centromere antibody
C. Anti-CCP antibody
D. Anti-Scl-70 (topoisomerase I) antibody
E. Anti-dsDNA antibody

3. A liver biopsy from a woman with acute liver failure in the third trimester of pregnancy is reviewed.

Which histologic finding is most characteristic of acute fatty liver of pregnancy?

A. Macrovesicular steatosis with Mallory-Denk bodies
B. Microvesicular steatosis
C. PAS-positive globules in periportal hepatocytes
D. Noncaseating granulomas
E. Periportal fibrin deposits and hemorrhage

4. A 26-year-old man has his third episode in 6 months of severe nausea, vomiting, and abdominal cramps. He spends hours in hot showers, which help. He has used cannabis daily for 5 years. Between episodes he is well.

Which of the following is the most effective long-term treatment?

A. Amitriptyline prophylaxis for migraine
B. Pyloric botulinum injection
C. A scopolamine patch before each episode
D. Stopping cannabis
E. Daily ondansetron

5. A student traces cholecalciferol from the skin to its active form.

What is the correct order?

A. Skin, then liver, then kidney
B. Kidney, then liver, then skin
C. Liver, then skin, then kidney
D. Skin, then kidney, then liver
E. Liver, then kidney, then skin

6. A 41-year-old woman with a positive H. pylori stool antigen test had anaphylaxis to penicillin as a child. Local clarithromycin resistance is low.

Which triple therapy regimen is most appropriate?

A. PPI + clarithromycin + metronidazole
B. PPI + amoxicillin + rifabutin
C. PPI alone for 8 weeks
D. PPI + amoxicillin + levofloxacin
E. PPI + amoxicillin + clarithromycin

7. A 29-year-old man with moderate-to-severe Crohn disease that has not responded to corticosteroids and azathioprine is to start infliximab.

Which test should be performed before starting therapy?

A. Serum ceruloplasmin
B. Thiopurine methyltransferase (TPMT) enzyme activity level
C. Interferon-gamma release assay (or tuberculin skin test)
D. Pulmonary function tests
E. Anti-Saccharomyces cerevisiae antibodies

8. A 34-year-old woman taking combined oral contraceptives, who has had three pregnancies, is found to have cholesterol gallstones.

How does estrogen promote cholesterol gallstone formation?

A. It increases cholesterol secretion into bile
B. It increases gallbladder contraction
C. It causes hemolysis
D. It increases bile salt synthesis
E. It deconjugates bilirubin

9. A 56-year-old woman has jaundice, dark urine, and pale stools.

Which finding best explains her dark urine?

A. Stercobilin reaching the urine
B. Conjugated bilirubin reaching the urine
C. Urobilin falling in the urine
D. Unconjugated bilirubin reaching the urine
E. Biliverdin reaching the urine

10. A researcher wants to study the cells that build the mucosal barrier in the human stomach.

Which cells form this barrier?

A. Enterochromaffin-like and chief cells
B. Parietal cells and their canaliculi
C. Foveolar and mucous neck cells
D. G cells and D cells of the antrum
E. Chief cells at the gland base

11. A 26-year-old woman at 30 weeks' gestation develops fever, nausea, and jaundice 5 weeks after returning from a trip where she drank untreated water. Within days she becomes confused, with an INR of 2.8.

Which virus most likely caused her illness?

A. Hepatitis D virus
B. Hepatitis B virus
C. Hepatitis A virus
D. Hepatitis E virus
E. Hepatitis C virus

12. Several children at a childcare center develop fever, abdominal cramps, and small-volume bloody, mucoid stools. The outbreak spreads person to person despite no shared food source. Stool culture grows a non-motile, lactose-nonfermenting gram-negative rod that does not produce H2S.

Which organism is most likely responsible?

A. Shigella
B. Enterotoxigenic E. coli
C. Bacillus cereus
D. Norovirus
E. Salmonella

13. A 30-year-old man has intense sympathetic activation immediately after a road crash.

Which effect on the stomach is expected, and through which fibers?

A. Stimulation through efferent CN X fibers
B. Inhibition through efferent CN X fibers
C. Stimulation through sympathetic fibers
D. No effect, as only CN X fibers reach the stomach
E. Inhibition through thoracolumbar sympathetic fibers

14. A 63-year-old man has painless jaundice. Imaging shows strictures of the bile ducts and a diffusely enlarged, "sausage-shaped" pancreas. Serum IgG4 is markedly elevated, and no mass is seen.

Which is the most appropriate treatment?

A. Pancreaticoduodenectomy
B. Liver transplant
C. Ursodeoxycholic acid alone
D. Broad-spectrum ABX
E. Corticosteroids

15. A 66-year-old man with painless jaundice has a CT showing a 2.5-cm mass in the head of the pancreas, with dilation of both the common bile duct and the pancreatic duct. There is no vascular invasion or metastasis.

Which is the most appropriate definitive treatment?

A. Pancreaticoduodenectomy (Whipple procedure)
B. Observation with repeat CT in 3 months
C. Cholecystectomy
D. Distal pancreatectomy
E. ERCP stent alone as definitive treatment

16. A 64-year-old woman with autoimmune gastritis has megaloblastic anemia, low serum B12, and reduced vibration sense in her feet.

Which is the most appropriate treatment?

A. Oral folic acid alone
B. Intramuscular vitamin B12
C. Oral iron
D. Proton pump inhibitor
E. H. pylori eradication therapy

17. A 56-year-old woman is about to start highly emetogenic chemotherapy. During her first cycle she received ondansetron alone, and her vomiting settled after the first day but returned on days 2–4.

Which drug added to her regimen best targets the vomiting on days 2–4?

A. Aprepitant
B. Scopolamine patch
C. Granisetron
D. Dimenhydrinate
E. Hyoscine butylbromide

18. A 45-year-old woman has heartburn that mainly wakes her at night. She is started on famotidine.

How does this drug reduce her symptoms?

A. It irreversibly binds the pump after acid activation
B. It competitively blocks histamine, decreasing cAMP
C. It polymerizes into a paste over the injured mucosa
D. It mimics PG, increasing mucus and decreasing HCl
E. It neutralizes HCl already present in the gastric lumen

19. A 54-year-old man with chronic hepatitis C has progressive fibrosis on biopsy, with dense deposits of collagen I.

Which process produces this collagen?

A. Hepatocytes secreting collagen into the bile canaliculi
B. Kupffer cells becoming fibroblasts in the sinusoids
C. Endothelial cells forming scar in the central vein
D. Bile duct cells secreting collagen into the portal vein
E. Stellate cells becoming myofibroblasts

20. A 48-year-old woman has 18 hours of constant right upper quadrant pain radiating to her right shoulder, with nausea. Temperature is 38.4°C. She stops breathing in when the right upper quadrant is palpated. WBC is 15800/μL; bilirubin, ALP, and lipase are normal. Ultrasound shows gallstones, a 5-mm gallbladder wall, and fluid around the gallbladder.

Which is the most appropriate definitive management?

A. Oral ursodeoxycholic acid
B. IV ABX and laparoscopic cholecystectomy during this admission
C. ERCP
D. Outpatient follow-up in 6 weeks
E. Percutaneous cholecystostomy as first-line in all patients

21. A 50-year-old woman has sudden severe epigastric pain radiating to her back with vomiting. Lipase is 9 times the upper limit of normal, ALT is 320 U/L, and ALP is raised. She drinks alcohol rarely. Ultrasound shows gallstones.

Where is the obstructing stone most likely located?

A. Gallbladder fundus
B. Cystic duct
C. Right hepatic duct at the porta hepatis
D. Common hepatic duct near the hilum
E. Hepatopancreatic ampulla (of Vater)

22. After surgery, a patient vomits after a small dose of IV morphine, yet another patient on a much higher fentanyl infusion has no nausea.

Which explanation fits these observations?

A. Morphine acts on vestibular H1 receptors, while fentanyl does not
B. Fentanyl causes nausea only after 24 hours, by releasing substance P
C. Fentanyl blocks D2 receptors in the CTZ
D. High doses reach μ receptors in the NTS behind the blood-brain barrier, which suppress vomiting
E. Morphine releases 5-HT from the gut wall, while fentanyl stays in the blood

23. A 78-year-old man has sudden, painless passage of a large volume of bright red blood per rectum. He is otherwise well, with a soft, nontender abdomen.

Which is the most likely cause?

A. Anal fissure
B. Ischemic colitis
C. Diverticulosis
D. Ulcerative colitis
E. Internal hemorrhoids

24. A 62-year-old obese man with 20 years of poorly controlled GERD undergoes endoscopy. Biopsies from salmon-colored mucosa above the GE junction show columnar epithelium with goblet cells.

This patient is at greatest risk of developing which of the following?

A. Squamous cell carcinoma of the upper esophagus
B. Adenocarcinoma of the distal esophagus
C. Gastric MALT lymphoma
D. Zenker diverticulum
E. Esophageal leiomyoma

25. A 23-year-old woman who smokes has crampy right lower quadrant pain, non-bloody diarrhea, and weight loss. She also has a perianal fistula. Colonoscopy shows patchy ulcers in the terminal ileum and ascending colon separated by normal mucosa, and the rectum is spared.

Which biopsy finding is most likely?

A. PAS-positive foamy macrophages
B. Caseating granulomas
C. Noncaseating granulomas with transmural inflammation
D. Crypt abscesses limited to the mucosa
E. Villous atrophy with intraepithelial lymphocytes

26. A 55-year-old man with chronic GERD has endoscopy showing Barrett esophagus without dysplasia. He is started on a daily PPI.

Which is the most appropriate surveillance plan?

A. Esophagectomy
B. No further endoscopy is needed
C. Repeat endoscopy every 3 months
D. Repeat endoscopy every 3–5 years
E. Annual barium swallow

27. A 19-year-old woman has had dull, poorly localized pain around her umbilicus since last night. This morning the pain moved to her right lower quadrant and became sharp, and she has vomited once. She is tender a third of the way from the right ASIS to the umbilicus.

Why did the character of her pain change?

A. Inflammation spread from the visceral wall to the parietal peritoneum
B. The inflamed organ began to bleed into the retroperitoneum
C. A ureteric stone moved from the kidney to the pelvic brim
D. Pain began to be referred through a shared spinal level
E. The organ perforated, releasing free air under the diaphragm

28. A 28-year-old woman at 20 weeks' gestation has 1 day of abdominal pain, now worst in the right side of the abdomen, with nausea and a temperature of 38.0°C. WBC is 15000/μL.

Which is the most appropriate initial imaging?

A. No imaging; observe for 48 hours and repeat the white cell count
B. Graded-compression ultrasound, followed by MRI if inconclusive
C. Barium enema
D. CT AP with contrast
E. Abdominal x-ray

29. A 21-year-old man has mild jaundice after a two-day fast, with raised unconjugated bilirubin and normal LFTs otherwise.

Which is the most likely diagnosis?

A. Crigler-Najjar syndrome type I
B. Choledocholithiasis
C. Rotor syndrome
D. Gilbert syndrome
E. Dubin-Johnson syndrome

30. A 38-year-old woman has a 3-cm liver lesion found by chance on ultrasound. Contrast MRI shows peripheral nodular enhancement that gradually fills in toward the center. She has no symptoms or liver disease.

Which is the most appropriate management?

A. Percutaneous biopsy
B. Reassurance; no biopsy or treatment needed
C. Surgical resection
D. Transarterial chemoembolization
E. Stop any oral contraceptives and repeat imaging in 1 month

31. A 28-year-old man wants fast relief of occasional heartburn after large meals.

Which drug class acts on HCl already in the lumen?

A. H2R antagonists
B. Misoprostol
C. PPIs
D. Sucralfate
E. Antacids

32. A 74-year-old woman with A-fib who is not anticoagulated has sudden, severe periumbilical pain that began 3 hours ago. She is writhing in pain, but her abdomen is soft with only mild tenderness. Serum lactate is elevated.

Which is the most appropriate diagnostic test?

A. Colonoscopy with mucosal biopsy
B. Barium enema with air contrast
C. Plain abdominal x-ray
D. Abdominal ultrasound
E. CT angiography of the abdomen

33. A 27-year-old man has bloating, gas, and watery diarrhea within hours of drinking milk. His symptoms stop when he avoids dairy.

Which treatment most directly addresses the deficient step?

A. A gluten-free diet
B. Lactase replacement
C. IM B12 replacement
D. Oral vancomycin
E. Pancreatic enzyme replacement

34. A 20-year-old student notices yellow eyes after 2 days of fasting during a bout of gastroenteritis. Total bilirubin is 48 μmol/L with direct bilirubin 4 μmol/L. AST, ALT, ALP, FBC, and blood film are normal.

Which of the following is the most appropriate management?

A. Ursodeoxycholic acid
B. Liver biopsy
C. MRCP to look for a bile duct stone
D. Phototherapy
E. No treatment

35. A 58-year-old man with alcohol-related cirrhosis and ascites has fever, diffuse abdominal tenderness, and new confusion. Diagnostic paracentesis shows an ascitic neutrophil count of 520/μL.

Which is the most appropriate treatment?

A. IV third-generation cephalosporin plus IV albumin
B. Large-volume paracentesis with albumin, without ABX
C. Oral rifaximin alone
D. Emergency laparotomy
E. Increase the diuretic dose

36. A 29-year-old woman taking oral contraceptives is diagnosed with Budd-Chiari syndrome on Doppler ultrasound. She has mild ascites, a normal mental status, INR 1.3 and no evidence of cirrhosis.

Which is the most appropriate initial treatment?

A. Immediate liver transplantation
B. Observation with repeat ultrasound in 6 months
C. Transjugular intrahepatic portosystemic shunt (TIPS) before anticoagulation
D. Anticoagulation and stopping the oral contraceptive
E. Large-volume paracentesis alone

37. A 38-year-old woman with poorly controlled type 2 diabetes develops severe epigastric pain radiating to her back. Her blood sample looks milky, and lipase is markedly raised. She does not drink alcohol, and ultrasound shows no gallstones.

Which laboratory finding most likely explains her pancreatitis?

A. Elevated serum ferritin
B. Positive anti-mitochondrial antibody
C. Serum triglycerides of 2400 mg/dL
D. Serum calcium of 8.0 mg/dL
E. Elevated serum IgG4

38. A physiologist notes that toxins in the blood, gastric irritation, motion, and fear can all cause vomiting.

Where do these different inputs converge?

A. The vestibular nuclei
B. The dorsal motor nucleus of the vagus
C. The nucleus tractus solitarius
D. The hypothalamus
E. The nucleus ambiguus

39. An 81-year-old woman with long-standing gallstones has chronic watery diarrhea and weight loss. Imaging shows air in the biliary tree and an abnormal connection between the gallbladder and the hepatic flexure of the colon.

Which best explains her diarrhea?

A. Lactose intolerance
B. Pancreatic enzyme deficiency
C. Bacterial overgrowth in the stomach
D. Bile salts entering the colon directly
E. Small bowel obstruction

40. A 19-year-old man has a 6-month history of hand tremor, slurred speech, and new irritability. His AST and ALT are mildly elevated. Slit-lamp examination shows a golden-brown ring at the edge of each cornea. Serum ceruloplasmin is low.

A defect in which process best explains his condition?

A. Intestinal iron absorption
B. Secretion of alpha-1 antitrypsin from hepatocytes
C. Conjugation of bilirubin
D. Hepatocyte copper excretion into bile
E. Hepatic uptake of LDL cholesterol

41. A 71-year-old man who smokes has 3 months of progressive dysphagia to solids and liquids and has lost 11 kg. Barium swallow shows a "bird's beak" narrowing at the gastroesophageal junction.

Which is the most appropriate next step?

A. Trial of a calcium channel blocker
B. Botulinum toxin injection
C. Pneumatic dilation
D. Esophageal manometry
E. Upper endoscopy

42. A 31-year-old man has chronic diarrhea, bloating, and weight loss. He eats a regular diet that includes bread and pasta.

Which is the best initial test for the most likely diagnosis?

A. Hydrogen breath test
B. Upper endoscopy with duodenal biopsy
C. IgA anti-tTG antibodies with total serum IgA
D. Fecal elastase
E. Stool ova and parasites on three separate samples

43. A 48-year-old man is diagnosed with HFE hereditary hemochromatosis. His hemoglobin is 15.2 g/dL, ferritin is 1600 ng/mL, and he has no cardiac disease.

Which is the most appropriate treatment?

A. Therapeutic phlebotomy
B. Low-iron diet alone
C. Liver transplant
D. D-penicillamine
E. Oral deferasirox

44. A 66-year-old man with rheumatoid arthritis needs long-term NSAID therapy and has a history of a gastric ulcer. He is started on a drug that is a prostaglandin E1 analog to protect the gastric mucosa.

This drug is contraindicated in which group of patients?

A. Patients with asthma
B. Patients with renal impairment
C. Pregnant patients
D. Patients taking warfarin
E. Patients with hypothyroidism

45. A 61-year-old man using an inhaled corticosteroid for COPD and recently treated with broad-spectrum ABX has painful swallowing. Endoscopy shows adherent white plaques along the esophagus.

Which finding would most likely be seen on a biopsy of the plaques?

A. Owl's eye inclusions in endothelial cells
B. More than 15 eosinophils per high-power field
C. Pseudohyphae and budding yeast
D. Goblet cells in columnar epithelium
E. Multinucleated giant cells

46. A researcher follows labeled copper from the duodenal lumen across an enterocyte and into the blood.

Which pair of transporters does it use?

A. DMT1, then ferroportin
B. SGLT1, then GLUT2
C. CTR1, then ATP7A
D. PepT1, then ATP7B
E. ATP7B, then CTR1

47. A 70-year-old woman with vitamin D deficiency has low serum calcium despite a normal calcium intake.

Which best explains her low calcium?

A. Decreased absorption in the duodenum
B. Decreased absorption in the ileum
C. Increased uptake through DMT1
D. Decreased binding to haptocorrin
E. Increased breakdown by trypsin

48. A 19-year-old woman is brought in 6 hours after taking 30 tablets of 500 mg paracetamol. She has mild nausea but is otherwise well. Her paracetamol level is above the treatment line on the nomogram.

The antidote she should receive works mainly by which mechanism?

A. Enhancing glucuronidation of paracetamol
B. Inhibiting CYP2E1
C. Binding paracetamol in the gut
D. Increasing renal excretion of paracetamol
E. Replenishing glutathione to neutralize NAPQI

49. A 33-year-old woman with diarrhea-predominant IBS and persistent abdominal pain is started on low-dose amitriptyline at night.

Which adverse effects is she most likely to experience?

A. Diarrhea and hypersalivation
B. Hyperkalemia and gynecomastia
C. Bradycardia and miosis
D. Dry mouth and constipation
E. Weight loss and insomnia

50. A 31-year-old man is brought in after a motorcycle collision. He has right upper quadrant tenderness and abrasions over the right lower ribs. Blood pressure is 128/76 mm Hg and heart rate 88/min, and they remain stable after 1 L of crystalloid.

Which is the most appropriate next step?

A. Upper endoscopy
B. Diagnostic peritoneal lavage
C. Discharge with outpatient follow-up
D. Emergency laparotomy
E. CT abdomen with IV contrast

Test A answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	**14** A B C D E	**27** A B C D E	**40** A B C D E
2 A B C D E	**15** A B C D E	**28** A B C D E	**41** A B C D E
3 A B C D E	**16** A B C D E	**29** A B C D E	**42** A B C D E
4 A B C D E	**17** A B C D E	**30** A B C D E	**43** A B C D E
5 A B C D E	**18** A B C D E	**31** A B C D E	**44** A B C D E
6 A B C D E	**19** A B C D E	**32** A B C D E	**45** A B C D E
7 A B C D E	**20** A B C D E	**33** A B C D E	**46** A B C D E
8 A B C D E	**21** A B C D E	**34** A B C D E	**47** A B C D E
9 A B C D E	**22** A B C D E	**35** A B C D E	**48** A B C D E
10 A B C D E	**23** A B C D E	**36** A B C D E	**49** A B C D E
11 A B C D E	**24** A B C D E	**37** A B C D E	**50** A B C D E
12 A B C D E	**25** A B C D E	**38** A B C D E	
13 A B C D E	**26** A B C D E	**39** A B C D E	

Test B

50 questions · 60 minutes

1. A 24-year-old woman with Wilson disease has finished initial copper chelation and needs maintenance treatment.

Which drug works by blocking intestinal copper absorption?

A. Zinc
B. Deferoxamine
C. Cholestyramine
D. D-penicillamine
E. Trientine

2. A 24-year-old man with hemolytic anemia has jaundice. His urine and stool are normal in color.

Which type of hyperbilirubinemia is most likely?

A. Unconjugated, as it is water soluble
B. Unconjugated, as it is bound to albumin
C. Conjugated, as MRP2 is defective
D. Conjugated, as bile is obstructed
E. Conjugated, as it is water soluble

3. A 34-year-old man with ulcerative colitis has an elevated ALP of 480 U/L on routine bloods, with normal bilirubin. Ultrasound shows no gallstones. He is p-ANCA positive and AMA negative.

Which test is most appropriate to confirm the diagnosis?

A. Serum IgG4 alone
B. Liver biopsy
C. MRCP
D. HIDA scan
E. Repeat ultrasound in 6 months

4. A pharmacology student asks why the area postrema can respond to toxins in the blood when most of the brainstem cannot.

Which feature of the area postrema explains this?

A. Its ependyma actively pumps toxins in from the CSF
B. It lies in the floor of the third ventricle, beside the hypothalamus
C. It is bathed in lymph drained from the abdominal organs
D. Its capillaries lack tight junctions, so it sits outside the blood-brain barrier
E. Its neurons send axons out through the vagus nerve to the gut

5. A 52-year-old man is admitted with acute pancreatitis. Heart rate is 112/min, BUN is rising, and urine output is low.

Which is the most important initial treatment?

A. Total parenteral nutrition
B. Goal-directed IV fluids such as Hartmann solution
C. Prophylactic IV carbapenem to prevent infected necrosis
D. Urgent contrast CT
E. Emergency necrosectomy

6. A 58-year-old man with cirrhosis has breast enlargement, small testes, reddened palms, and many small, spider-like blood vessels on his chest.

Which mechanism best explains these findings?

A. Decreased albumin synthesis by the failing liver
B. Vitamin K deficiency from reduced bile salt secretion
C. Increased ammonia from reduced hepatic clearance
D. Portal hypertension
E. Increased estrogen from reduced hepatic clearance

7. A 50-year-old woman with a duodenal ulcer and confirmed H. pylori infection needs a drug that coats the ulcer base and also kills the organism.

Which drug is most appropriate?

A. Aluminum hydroxide gel
B. Sucralfate
C. Bismuth subcitrate
D. Famotidine
E. Misoprostol

8. A 57-year-old man with untreated hereditary hemochromatosis develops fever, hypotension, and hemorrhagic bullae on his legs 1 day after eating raw oysters.

Which organism is most likely responsible?

A. Staphylococcus aureus
B. Vibrio cholerae
C. Vibrio vulnificus
D. Bacillus cereus
E. Listeria monocytogenes

9. A 45-year-old woman with diabetic gastroparesis has early satiety and vomiting of undigested food. She starts a drug that relieves her nausea and also speeds gastric emptying.

Which pair of actions best explains both effects?

A. 5-HT3 blockade on vagal afferents and muscarinic blockade in the gut
B. H1 blockade in the brain and 5-HT3 blockade in the CTZ
C. μ-opioid stimulation in the NTS and D2 stimulation in the gut
D. D2 blockade in the CTZ and 5-HT4 stimulation in the gut
E. NK1 blockade in the NTS and H1 blockade in the brain

10. A 26-year-old woman develops watery diarrhea 3 days after arriving in Mexico. She has 4 loose stools a day, no fever and no blood in the stool, and is drinking well.

Which is the most appropriate management?

A. Stool ova and parasites, then empirical metronidazole
B. Oral rehydration, with loperamide if needed
C. IV ciprofloxacin and admission
D. Oral vancomycin
E. Urgent colonoscopy

11. A 24-year-old man with allergic rhinitis is diagnosed with eosinophilic esophagitis after an episode of food impaction. He asks how to address the underlying cause of his condition.

Which management targets the trigger of this condition?

A. Pneumatic dilation of the LES
B. Elimination of food allergens from the diet
C. Long-term oral fluconazole
D. Laparoscopic Nissen fundoplication for reflux
E. Stopping all oral medications for 2 weeks

12. A 69-year-old man in the ICU with cardiogenic shock is receiving high-dose noradrenaline. He develops abdominal distention and rising lactate. CT angiography shows diffuse narrowing of the mesenteric arteries without an embolus or thrombus.

Which is the most likely diagnosis?

A. Mesenteric venous thrombosis
B. Non-occlusive mesenteric ischemia
C. Chronic mesenteric ischemia
D. SMA embolus
E. Volvulus

13. A 30-year-old man has bloody diarrhea with tenesmus. Colonoscopy shows continuous inflammation from the rectum to the splenic flexure. His alkaline phosphatase is elevated, and MRCP shows multifocal strictures and dilations ("beading") of the intra- and extrahepatic bile ducts.

Besides colorectal cancer, this patient is at greatest risk for which malignancy?

A. Cholangiocarcinoma
B. Gastric MALT lymphoma
C. Small-bowel lymphoma
D. Hepatocellular carcinoma
E. Pancreatic adenocarcinoma

14. A 58-year-old man has had untreated achalasia for 25 years, with a markedly dilated esophagus full of retained food.

He is at increased risk of which malignancy?

A. Leiomyosarcoma of the esophageal muscle wall
B. Hepatocellular carcinoma
C. Squamous cell carcinoma of the esophagus
D. Adenocarcinoma of the proximal stomach
E. Gastric MALT lymphoma

15. A 56-year-old man has 2 days of right upper quadrant pain, jaundice, dark urine, and itching. He is afebrile. Bilirubin is 5.8 mg/dL and ALP is 410 U/L, with mildly raised AST/ALT and normal lipase. Ultrasound shows gallstones and a common bile duct dilated to 10 mm.

Which is the most appropriate next step?

A. Laparoscopic cholecystectomy alone
B. ERCP with stone extraction
C. CT-guided liver biopsy
D. HIDA scan
E. Repeat LFTs in 2 weeks

16. A 61-year-old man has had heartburn for years. Over the past 2 months he has had difficulty swallowing solid food and has lost 6 kg. His hemoglobin is 10.8 g/dL with a low MCV.

Which is the most appropriate next step?

A. 24-hour pH monitoring
B. H. pylori stool antigen test
C. Upper endoscopy with biopsy
D. 8-week trial of a PPI
E. Esophageal manometry

17. A 79-year-old woman has small bowel obstruction. CT shows air in the biliary tree, dilated small bowel, and a 3.5-cm gallstone in the distal ileum. She is resuscitated.

Which is the most appropriate surgical procedure?

A. Emergency cholecystectomy and fistula repair as the first step
B. Colonoscopic stone removal
C. Enterolithotomy to remove the stone from the bowel
D. Nasogastric decompression alone
E. Right hemicolectomy

18. A patient has a stone in the common bile duct and develops an infection of the obstructed bile.

Which combination of findings makes up Charcot triad?

A. Painless jaundice, palpable gallbladder, weight loss
B. Right upper quadrant pain, Murphy sign, fever
C. Jaundice, pruritus, pale stools
D. Fever, right upper quadrant pain, jaundice
E. Fever, hypotension, confusion

19. A 34-year-old man with HIV who stopped antiretroviral therapy a year ago presents with 2 weeks of painful swallowing. His CD4 count is 62/μL. Examination shows white plaques on the buccal mucosa that scrape off, leaving an erythematous base. He is hemodynamically stable and can tolerate oral fluids.

Which of the following is the most appropriate next step in management?

A. Empiric oral fluconazole
B. Oral acyclovir
C. IV ganciclovir
D. High-dose oral omeprazole
E. Upper endoscopy with biopsy before any treatment

20. A student chews a piece of bread and notices it begins to taste sweet. Once the bolus has mixed with gastric juice, this starch digestion stops.

Which best explains why it stops?

A. It needs enterokinase, found in the duodenum
B. Gastric lipase competes for the same starch
C. Pepsin converts it to an inactive zymogen
D. Stomach acid stops salivary amylase
E. Bile salts bind it in the gastric lumen

21. A 59-year-old man with a duodenal ulcer takes high-dose omeprazole. Sucralfate is added, but it provides little extra benefit.

Which best explains this?

A. It needs a high pH to bind the ulcer
B. It is neutralized by bicarbonate
C. It acts on H2 receptors, now blocked
D. Omeprazole blocks its target
E. It needs acid to polymerize into a paste

22. A 28-year-old woman who has taken oral contraceptives for 10 years presents with sudden right upper quadrant pain and hypotension. CT shows a 7-cm liver mass with surrounding hemorrhage.

Which is the most likely underlying lesion?

A. Angiosarcoma
B. Focal nodular hyperplasia
C. Hemangioma
D. Hepatocellular carcinoma
E. Hepatic adenoma

23. A 35-year-old man is admitted with full-thickness burns over 45% of his body surface area. On hospital day 3, he has an episode of coffee-ground emesis.

Which mechanism best explains this complication?

A. Urease-mediated mucosal injury
B. Inhibition of prostaglandin synthesis
C. Increased vagal stimulation of parietal cells
D. Gastrin-secreting tumor
E. Hypovolemia leading to gastric mucosal ischemia

24. A 61-year-old woman with vitiligo and Hashimoto thyroiditis has fatigue and numbness in her feet. Hemoglobin is 9.1 g/dL with an MCV of 112 fL. Antibodies to parietal cells and intrinsic factor are positive.

Which region of the stomach is most likely to show atrophy?

A. Body and fundus
B. Pylorus
C. Antrum
D. Cardia only
E. Duodenal bulb

25. A student traces bilirubin from heme to the blood.

Which enzyme forms unconjugated bilirubin?

A. Biliverdin reductase
B. UGT
C. Glutathione reductase
D. Heme oxygenase
E. Alcohol dehydrogenase

26. A 31-year-old woman at 36 weeks' gestation has nausea, abdominal pain, and jaundice. Blood pressure is 128/80 mm Hg. Two possible diagnoses are being considered: acute fatty liver of pregnancy and HELLP syndrome.

Which finding most favors acute fatty liver of pregnancy?

A. Platelet count of 60000/μL with schistocytes
B. Raised LDH
C. Proteinuria
D. Blood pressure over 160/110 mm Hg
E. Hypoglycemia with prolonged INR

27. A 22-year-old man has 18 hours of abdominal pain that began around the umbilicus and moved to the right lower quadrant, with anorexia and a low-grade fever. Lying on his left side, he has pain when his right hip is passively extended.

This finding suggests the inflamed appendix lies in which position?

A. Anterior, just under McBurney point
B. Left lower quadrant
C. Pelvic, against the obturator internus
D. Retrocecal, against the iliopsoas muscle
E. Subhepatic

28. A 23-year-old woman with acute liver failure after a paracetamol overdose becomes sweaty and less responsive in the ICU. Her INR is 4.2.

Which should be checked immediately?

A. Thyroid-stimulating hormone
B. Serum lipase
C. Ceruloplasmin
D. Serum ferritin
E. Blood glucose

29. A 19-year-old man is tackled during a football match and develops right upper quadrant pain. He also complains of pain at the tip of his right shoulder.

The shoulder pain is referred through which nerve?

A. Vagus nerve
B. Intercostal nerve
C. Phrenic nerve
D. Greater splanchnic nerve
E. Long thoracic nerve

30. A 63-year-old woman who takes naproxen daily develops sudden, severe epigastric pain that spreads across the abdomen. She is tachycardic, and the abdomen is rigid with rebound tenderness. An upright chest x-ray shows air under the right hemidiaphragm.

In addition to IV fluids, IV PPI, and ABX, which is the most appropriate next step?

A. Emergency surgical repair
B. Barium swallow to locate the perforation
C. Urgent upper endoscopy
D. Nasogastric tube and serial abdominal examinations only
E. H. pylori testing and observation

31. A 33-year-old woman has had 10 days of intractable hiccups, nausea, and vomiting. Endoscopy and abdominal imaging are normal. Six months ago she had severe optic neuritis that left poor vision in one eye.

Which test is most likely to explain her symptoms?

A. β-hCG
B. Gastric emptying study
C. Serum AQP4 antibody
D. Serum lipase
E. Urine cannabinoid screen

32. A 66-year-old woman has 2 days of left lower quadrant pain and fever. Her white cell count is 15.8 × 10^9/L, and CT AP shows sigmoid wall thickening with fat stranding and no abscess. She asks whether she should have a colonoscopy now.

Which reply is correct?

A. Yes, today, to take biopsies of the inflamed segment
B. No, colonoscopy is avoided during acute diverticulitis
C. Yes, because colonoscopy is the first test for LLQ pain
D. No, because she is past the age for colonoscopy screening
E. No, because CT has excluded any colonic disease

33. A 78-year-old woman with a history of gallstones has 3 days of colicky abdominal pain, distention, vomiting, and no passage of stool or gas. CT shows air in the biliary tree, dilated small bowel loops, and a 3-cm calcified stone in the right lower quadrant.

Where is the stone most likely lodged?

A. Duodenojejunal flexure
B. Pylorus
C. Sigmoid colon
D. Ileocecal valve
E. Ampulla of Vater

34. A 42-year-old woman with obesity has had three episodes of epigastric pain radiating to her back, each starting about an hour after a large fried meal and settling within 3 hours. She is pain-free today, afebrile, with a soft, nontender abdomen. LFTs and white cell count are normal. Ultrasound shows several mobile, shadowing gallstones and a normal gallbladder wall.

Which is the most appropriate management?

A. No treatment; reassure
B. IV ABX and admission for emergency surgery
C. Elective laparoscopic cholecystectomy
D. HIDA scan
E. ERCP with sphincterotomy

35. A 58-year-old man with celiac disease who has not followed a gluten-free diet presents with weight loss, abdominal pain, and a small bowel perforation. Pathology of the resected segment shows a T-cell malignancy.

Which is the most likely diagnosis?

A. Colorectal adenocarcinoma
B. Carcinoid tumor
C. Extranodal marginal zone (MALT) lymphoma
D. Burkitt lymphoma
E. Enteropathy-associated T-cell lymphoma

36. A 47-year-old man who drinks a bottle of vodka daily has fever, jaundice, and tender hepatomegaly after a week of heavy drinking. AST is 240 U/L, ALT is 95 U/L, GGT is elevated, and MCV is 104 fL.

Which is the most likely diagnosis?

A. Paracetamol toxicity
B. Ischemic hepatitis
C. Acute viral hepatitis A
D. Alcoholic hepatitis
E. MASLD

37. A 27-year-old woman has 8 months of recurrent lower abdominal cramping that improves after bowel movements, with alternating loose and hard stools. She has no weight loss, bleeding, or nocturnal symptoms. FBC, CRP, fecal calprotectin, and celiac serology are normal.

Which is the most appropriate next step?

A. CT AP
B. Reassurance and a trial of a low-FODMAP diet
C. Empiric metronidazole for small bowel overgrowth
D. Start oral prednisone for presumed IBD
E. Colonoscopy

38. A 25-year-old woman eats table sugar. Enzymes on the brush border of her enterocytes digest it before it can be absorbed.

Which enzyme acts, and what does it form?

A. Sucrase, forming glucose and galactose
B. Lactase, forming glucose and galactose
C. Lactase, forming glucose alone
D. Maltase, forming glucose and fructose
E. Sucrase, forming glucose and fructose

39. A baby is born to a mother who is HBsAg positive. The baby is healthy, at term, and weighs 3.4 kg.

Which is the most appropriate management for the newborn?

A. HBIG only
B. Tenofovir for the newborn for 6 months
C. Hepatitis B vaccine at 2 months of age
D. Hepatitis B vaccine plus HBIG within 12 hours of birth
E. Test the newborn for HBsAg at birth and treat only if positive

40. A 35-year-old woman with biliary obstruction has a prolonged PT that corrects after parenteral vitamin K.

Which best explains her vitamin deficiency?

A. Increased hepcidin degrading vitamin K
B. Decreased SGLT1 uptake of vitamin K
C. Decreased GLUT5 transport of vitamin K
D. Decreased IF binding of vitamin K
E. Decreased micelles to carry vitamin K

41. A 49-year-old woman with PBC starts ursodeoxycholic acid.

How does this drug help her?

A. It blocks viral reverse transcriptase in the hepatocyte
B. It binds bile salts in the gut and blocks their reuptake
C. It inhibits CYP7A1, stopping synthesis of new bile salts
D. It suppresses T cells that are attacking her bile ducts
E. It displaces toxic bile salts and increases bile flow

42. A 19-year-old woman has taken a large paracetamol overdose, and NAC is started.

How does NAC protect her liver?

A. It replenishes glutathione
B. It induces CYP2E1 activity
C. It adds sulfate to NAPQI
D. It reverses warfarin at VKOR
E. It blocks UGT in hepatocytes

43. A 70-year-old man has a cardiac arrest. Biopsy shows necrosis around the central veins.

Why is this zone injured first?

A. Blood reaches it first, so it has the most toxins
B. Blood reaches it last, so it has the least oxygen
C. It lies beside the bile duct of the portal triad
D. It has no sinusoids to supply its hepatocytes
E. It holds most of the Kupffer cells in each lobule

44. A 72-year-old woman with long-standing pulmonary hypertension and severe tricuspid regurgitation has jugular venous distention, bilateral leg edema, and a tender, enlarged liver. Pressing on the liver increases her jugular venous pulsation. At autopsy, the cut surface of the liver shows dark red centrilobular areas alternating with pale tan areas.

Which mechanism best explains the liver findings?

A. Raised central venous pressure transmitted to the hepatic veins
B. Portal vein thrombosis
C. Fatty infiltration of hepatocytes
D. Thrombosis of the hepatic veins blocking outflow from the sinusoids
E. Occlusion of the hepatic artery

45. A 5-year-old boy had bloody diarrhea for 4 days after eating an undercooked hamburger. He is now pale with petechiae and decreased urine output. Bloods show hemoglobin 7.8 g/dL, platelets 42000/μL, creatinine 2.1 mg/dL and schistocytes on the blood smear.

Which intervention should be avoided?

A. Monitoring electrolytes and urine output
B. Careful IV fluid management
C. ABX therapy
D. Red blood cell transfusion if symptomatic
E. Dialysis if indicated

46. A 71-year-old man has aching in his hips and an enlarged skull on examination. Blood tests show ALP 410 IU/L with normal GGT, bilirubin, AST, and ALT.

What is the most likely source of the raised ALP?

A. Bile duct obstruction
B. Bone
C. Hepatic infiltration by lymphoma
D. Hepatic metastases
E. Primary biliary cholangitis

47. A 35-year-old woman eats a fatty meal. Her gallbladder contracts, pancreatic enzymes are released, and gastric emptying slows.

Which cell and hormone best explain these effects?

A. L cells secreting PYY
B. S cells secreting secretin
C. G cells secreting gastrin
D. I cells secreting CCK
E. K cells secreting GIP

48. A 77-year-old man is in the ICU on vasopressors for septic shock. Two days later he develops crampy left-sided abdominal pain followed by bloody diarrhea.

Which region of the colon is most vulnerable to this injury?

A. Terminal ileum
B. Mid-transverse colon
C. Cecum and ascending colon
D. Splenic flexure
E. Distal rectum

49. A 59-year-old woman being treated for diverticulitis has persistent fever and pain after 3 days of IV ABX. CT shows a 6-cm pericolic fluid collection with an enhancing rim and no free air.

Which is the most appropriate next step?

A. CT-guided percutaneous drainage plus ABX
B. Colonoscopy
C. Continue IV ABX alone and repeat CT in 1 week
D. Increase dietary fiber
E. Emergency colectomy with end colostomy

50. A 45-year-old man from rural Southeast Asia, who often eats raw freshwater fish, presents with painless jaundice and weight loss. Imaging shows a mass at the confluence of the right and left hepatic ducts.

Which organism most likely increased his risk for this tumor?

A. Clonorchis sinensis
B. Helicobacter pylori
C. Entamoeba histolytica
D. Hepatitis A virus
E. Schistosoma haematobium

Test B answer sheet

Name _____ Date _____ Score _____ / 50

1	A B C D E	**14**	A B C D E	**27**	A B C D E	**40**	A B C D E
2	A B C D E	**15**	A B C D E	**28**	A B C D E	**41**	A B C D E
3	A B C D E	**16**	A B C D E	**29**	A B C D E	**42**	A B C D E
4	A B C D E	**17**	A B C D E	**30**	A B C D E	**43**	A B C D E
5	A B C D E	**18**	A B C D E	**31**	A B C D E	**44**	A B C D E
6	A B C D E	**19**	A B C D E	**32**	A B C D E	**45**	A B C D E
7	A B C D E	**20**	A B C D E	**33**	A B C D E	**46**	A B C D E
8	A B C D E	**21**	A B C D E	**34**	A B C D E	**47**	A B C D E
9	A B C D E	**22**	A B C D E	**35**	A B C D E	**48**	A B C D E
10	A B C D E	**23**	A B C D E	**36**	A B C D E	**49**	A B C D E
11	A B C D E	**24**	A B C D E	**37**	A B C D E	**50**	A B C D E
12	A B C D E	**25**	A B C D E	**38**	A B C D E		
13	A B C D E	**26**	A B C D E	**39**	A B C D E		

Test C

50 questions · 60 minutes

1. A 68-year-old man with several prior episodes of sigmoid diverticulitis reports passing air bubbles in his urine and has had 3 urinary tract infections in 4 months.

Which is the most likely explanation for his symptoms?

A. Colovesical fistula
B. Urethral stricture
C. Emphysematous cystitis
D. Bladder exstrophy
E. Benign prostatic hyperplasia

2. A 41-year-old man with known chronic hepatitis B, who injects drugs, has a sudden flare of hepatitis with jaundice. HBsAg is positive, IgM anti-HBc is negative, and IgG anti-HBc and anti-HDV are positive.

Which statement about his new infection is most accurate?

A. It is an HDV superinfection and is likely to become chronic
B. It is an HDV coinfection and will likely resolve
C. It can be prevented by the hepatitis C vaccine
D. It is reactivation of HBV alone
E. It is acute hepatitis A

3. A 30-year-old man becomes nauseated whenever he reads in a moving car. His brother, who lost all labyrinthine function after an infection, never does.

Which input explains why the brother is protected?

A. Vagal afferents from the stomach
B. The vestibular system
C. Baroreceptors in the carotid sinus
D. The cortex and thalamus
E. The chemoreceptor trigger zone

4. A 26-year-old woman with Crohn disease develops severe epigastric pain radiating to the back 4 weeks after starting a new medication. Lipase is 8 times normal. She does not drink alcohol, ultrasound shows no gallstones, and triglycerides and calcium are normal.

Which medication is the most likely cause?

A. Azathioprine
B. Budesonide
C. Loperamide
D. Vitamin D
E. Infliximab

5. A 68-year-old woman needs long-term NSAIDs and is started on misoprostol to protect her stomach.

Which mechanism explains this protection?

A. It is an irreversible pump inhibitor that stops HCl at its source
B. It is a PG analog that decreases HCl and increases mucus
C. It is an antacid that neutralizes HCl within the gastric lumen
D. It is a competitive H2 blocker that decreases cAMP in parietal cells
E. It is a paste that binds proteins at the base of the ulcer

6. A 61-year-old man with HCV cirrhosis has a 3-cm liver lesion on surveillance ultrasound. Multiphase CT shows arterial-phase enhancement with washout in the portal venous phase. AFP is 420 ng/mL.

Which is the most appropriate next step to confirm the diagnosis?

A. PET scan
B. Percutaneous biopsy
C. Diagnostic laparoscopy
D. No further test is needed
E. Repeat ultrasound in 3 months

7. A researcher follows labeled iron from the duodenal lumen into the blood.

Which sequence of proteins does it use?

A. Ferroportin, DMT1, then transferrin
B. DMT1, ferroportin, then transferrin
C. SGLT1, GLUT2, then transcobalamin
D. CTR1, ATP7A, then ceruloplasmin
E. Transferrin, DMT1, then ferroportin

8. A 52-year-old man with chronic pancreatitis has bulky, greasy stools and weight loss. Fecal elastase is very low.

Which treatment best replaces the missing step?

A. Lactase replacement
B. Intrinsic factor tablets
C. Bile salt supplements
D. A gluten-free diet
E. Pancreatic enzyme replacement

9. A 50-year-old woman develops fever, eosinophilia, and severe hepatitis 1 week after her second general anesthetic in 3 months. Both used a halogenated inhaled anesthetic.

Which mechanism best explains her liver injury?

A. Copper accumulation
B. Hypoperfusion of the liver during repeated prolonged surgery
C. Bile duct injury during surgery
D. Excess NAPQI formation
E. Immune reaction to drug-modified liver proteins

10. A 24-year-old woman is stabbed in the right upper abdomen. Her blood pressure is 78/40 mm Hg and heart rate 136/min despite 2 units of blood. A FAST exam shows free fluid in Morison pouch.

Which is the most appropriate next step?

A. Repeat FAST in 30 minutes
B. CT abdomen with IV contrast
C. Angioembolization
D. Emergency laparotomy
E. Serial abdominal examinations

11. A newborn has severe unconjugated hyperbilirubinemia and develops kernicterus. UGT activity is absent.

Which is the most likely diagnosis?

A. Crigler-Najjar syndrome type I
B. Rotor syndrome
C. Crigler-Najjar syndrome type II
D. Gilbert syndrome
E. Dubin-Johnson syndrome

12. A student charts how much HCl each phase contributes to the secretion that follows a meal.

Which phase contributes the most HCl, and about how much?

A. Cephalic phase, about 60%
B. Gastric phase, about 30%
C. Gastric phase, about 60%
D. Intestinal phase, about 30%
E. Intestinal phase, about 60%

13. A 59-year-old woman recovered from ascending cholangitis after ERCP with removal of a common bile duct stone. Ultrasound shows more gallstones in her gallbladder.

Which is the most appropriate step to prevent recurrence?

A. Cholecystectomy
B. Repeat ERCP every 6 months
C. No further treatment
D. Long-term oral ABX
E. Ursodeoxycholic acid only

14. A 62-year-old woman has megaloblastic anemia. Autoantibodies have destroyed a cell population in the gastric body and fundus.

Which pair of products has been lost?

A. Pepsinogen and gastric lipase
B. Mucus alone
C. Histamine alone
D. Gastrin and somatostatin
E. Protons and IF

15. A 50-year-old man with severe pancreatic insufficiency is found to have low B12, and his IF is normal.

Which step of B12 absorption is most directly impaired?

A. Proteases freeing B12 in the duodenum
B. Absorption of the complex in the terminal ileum
C. Transport of B12 in blood by transcobalamin
D. Secretion of IF from gastric parietal cells
E. Binding of B12 to haptocorrin in the stomach

16. A 27-year-old woman with ulcerative colitis has a severe flare with 10 bloody stools per day. Her temperature is 39.0°C and heart rate 124/min, and her abdomen is distended and tender. Abdominal x-ray shows a transverse colon dilated to 8 cm.

Which medication should be avoided?

A. IV fluids
B. Broad-spectrum IV ABX
C. Subcutaneous heparin for DVT prophylaxis
D. IV methylprednisolone
E. Loperamide

17. An 86-year-old woman with achalasia and severe heart failure is not a candidate for dilation or surgery. She receives an endoscopic injection into the lower esophageal sphincter.

The injected agent works by which mechanism?

A. Stimulating muscarinic receptors
B. Blocking acetylcholine release from nerve terminals
C. Blocking L-type calcium channels in smooth muscle
D. Inhibiting the proton pump
E. Releasing nitric oxide

18. A 32-year-old woman meets diagnostic criteria for IBS. Her clinician explains that her abdominal pain partly reflects increased central amplification of signals from gut stretch receptors.

Which condition is she most likely to have as well?

A. Ulcerative colitis
B. Fibromyalgia
C. Primary biliary cholangitis
D. Celiac disease
E. Hemochromatosis

19. A 55-year-old woman with primary biliary cholangitis on ursodeoxycholic acid has severe itching that keeps her awake. She is started on a drug that binds bile salts in the intestine and prevents their reabsorption.

Which drug was started?

A. Orlistat
B. Polyethylene glycol
C. Ezetimibe
D. Cholestyramine
E. Octreotide

20. A physiology lecture explains why the liver tolerates occlusion of the hepatic artery better than many other organs tolerate loss of their arterial supply.

Which vessel provides most of the liver's blood flow?

A. Inferior vena cava
B. Splenic artery
C. Portal vein
D. Hepatic artery proper
E. Hepatic veins

21. After an emergency delivery for acute fatty liver of pregnancy, the mother asks why the condition happened.

A deficiency in the fetus of which enzyme is most strongly associated?

A. Glucose-6-phosphatase
B. Hexosaminidase A
C. Medium-chain acyl-CoA dehydrogenase (MCAD)
D. Ornithine transcarbamylase
E. Long-chain 3-hydroxyacyl-CoA dehydrogenase (LCHAD)

22. A 31-year-old woman at 9 weeks of pregnancy has vomited many times a day for a week and has lost 6% of her weight. Urine shows ketones.

Which acid-base disturbance is her vomiting most likely to cause?

A. High anion gap metabolic acidosis from lactate
B. Respiratory acidosis
C. Hypochloremic metabolic alkalosis
D. Hyperchloremic metabolic acidosis
E. Respiratory alkalosis

23. A 68-year-old man has tender hepatomegaly, ascites, bilateral leg edema, and distended neck veins that rise further with pressure over the liver. Bilirubin and AST/ALT are mildly raised.

Which is the most appropriate next test?

A. Anti-smooth muscle antibody
B. Echocardiography
C. Liver biopsy
D. Doppler ultrasound of the hepatic veins
E. MRCP

24. A 9-year-old girl has 1 day of periumbilical pain that has moved to the right lower quadrant, with vomiting and a temperature of 38.1°C. She is tender at McBurney point. WBC is 14500/μL.

Which is the most appropriate initial imaging test?

A. Barium enema
B. CT AP with contrast
C. Right lower quadrant ultrasound
D. Colonoscopy
E. Erect and supine abdominal x-ray

25. A 68-year-old man who smokes has 6 weeks of progressive jaundice, itching, and a 7-kg weight loss, with no abdominal pain. On examination, he has a smooth, nontender mass below the right costal margin that moves with respiration. ALP and conjugated bilirubin are markedly raised.

Which is the most likely cause?

A. Pancreatic head or other periampullary malignancy
B. Choledocholithiasis
C. Mirizzi syndrome from a stone in the cystic duct
D. Hepatitis A
E. Acute cholecystitis with a stone in the gallbladder neck

26. A 40-year-old woman who becomes seasick on every boat trip is going on a 3-day cruise next week. She wants something to prevent it.

Which option is most appropriate, and how does it work?

A. Oral metoclopramide, blocking D2 receptors and speeding gastric emptying
B. Transdermal scopolamine, blocking muscarinic input to the vestibular nuclei
C. Oral loratadine, blocking peripheral H1 receptors in the inner ear
D. Oral aprepitant, blocking substance P in the area postrema
E. Oral ondansetron, blocking 5-HT3 receptors on vagal afferents

27. Two patients have chronic H. pylori infection. One strain carries a gene encoding a protein that is injected into gastric epithelial cells by a type IV secretion system.

Strains with this virulence factor carry a higher risk of which outcome?

A. Gastric adenocarcinoma
B. Eosinophilic esophagitis
C. Pancreatic insufficiency
D. Celiac disease
E. Achalasia

28. A 74-year-old woman with long-standing gallstones has air in the biliary tree on CT. She has no history of biliary surgery or ERCP.

Which type of biliary fistula is most common?

A. Cholecystocolic
B. Cholecystogastric
C. Cholecystocutaneous
D. Cholecystoduodenal
E. Choledochoduodenal

29. A 45-year-old man's brother was recently diagnosed with HFE hereditary hemochromatosis. He feels well and asks to be tested.

Which is the best initial test?

A. Transferrin saturation and serum ferritin
B. Echocardiography
C. MRI of the liver
D. Liver biopsy with Prussian blue stain
E. Serum ceruloplasmin

30. A 34-year-old woman has her terminal ileum resected for Crohn disease and later has a smaller bile salt pool.

Which step in bile salt handling has been lost?

A. Activation of bile salts by enterokinase
B. Formation of micelles in the jejunum
C. Reabsorption into the portal vein
D. Packaging of bile salts in chylomicrons
E. Secretion of bile salts by the pancreas

31. A 48-year-old man is given a somatostatin analog during a research study of gastric function.

Which cells does somatostatin inhibit?

A. Chief and mucous neck cells, but not parietal cells
B. Foveolar, mucous neck, and D cells, but not G cells
C. S, I, and K cells of the duodenum and the jejunum
D. G, ECL, parietal, and chief cells
E. Only parietal cells, acting through their H2 receptors

32. A 19-year-old man with asthma and atopic dermatitis presents to the emergency department with a piece of steak stuck in his throat. He reports years of intermittent dysphagia to solids. Endoscopy shows concentric esophageal rings and linear furrows. Biopsy shows more than 15 eosinophils per high-power field.

Which of the following is the most likely diagnosis?

A. Candida esophagitis
B. Schatzki ring
C. Achalasia
D. Eosinophilic esophagitis
E. Gastroesophageal reflux disease

33. A 60-year-old man with alcohol-related cirrhosis has moderate ascites. He has stopped drinking. Creatinine and sodium are normal.

Which is the most appropriate initial treatment?

A. Fluid restriction to 500 mL per day, with sodium intake left unrestricted
B. Furosemide alone
C. Sodium restriction and spironolactone, with furosemide added as needed
D. Immediate TIPS
E. Weekly large-volume paracentesis

34. A researcher follows labeled fructose from the lumen of the jejunum into the blood.

Which route does it take?

A. SGLT1 into the cell, then GLUT5 into the portal vein
B. Micelles into the cell, then chylomicrons into a lacteal
C. PepT1 into the cell, then GLUT2 into a nearby lacteal
D. GLUT2 into the cell, then SGLT1 into a nearby lacteal
E. GLUT5 into the cell, then GLUT2 into the portal vein

35. A 35-year-old woman is admitted 2 days after a paracetamol overdose. Over the next 3 days her ALT falls from 9800 to 2100 IU/L, while her bilirubin rises from 40 to 120 µmol/L and her INR from 2.1 to 4.6. She is becoming drowsy.

What do these trends indicate?

A. Vitamin K deficiency from poor intake
B. Recovery, since the hepatocyte injury is settling
C. Bile duct obstruction from swollen hepatocytes
D. A failing liver with fewer hepatocytes left to injure
E. Hemolysis from the paracetamol metabolite

36. A 66-year-old man in the ICU after major burns, on TPN and vasopressors, develops fever and leukocytosis. Ultrasound shows a distended gallbladder with a thickened wall and fluid around it, but no gallstones. He is too unstable for surgery.

Which is the most appropriate management?

A. Oral ursodeoxycholic acid
B. ERCP with sphincterotomy
C. IV ABX and percutaneous cholecystostomy
D. Observation only
E. Elective cholecystectomy in 6 weeks

37. A 58-year-old man had a cholecystectomy 6 years ago. He now has jaundice from a stone in the common bile duct. The stone is soft and brown, and bile culture grows E. coli.

Which best explains how this stone formed?

A. Calcium oxalate precipitation
B. Excess bile salt secretion
C. Cholesterol supersaturation in the gallbladder
D. Chronic hemolysis
E. Bacterial deconjugation of bilirubin within the bile duct

38. A 66-year-old man is in the ICU on mechanical ventilation for 4 days after a severe head injury. His platelet count is 48000/µL.

Which intervention best reduces his risk of stress-related gastric bleeding?

A. No intervention is needed
B. H. pylori testing
C. Routine daily upper endoscopy
D. Prophylactic misoprostol
E. Prophylactic PPI or H2 blocker

39. A student traces the urea cycle to its final step.

Which enzyme releases urea, and what does it regenerate?

A. Arginase, regenerating citrulline
B. Argininosuccinate lyase, regenerating ornithine
C. Arginase, regenerating ornithine
D. Ornithine transcarbamylase, regenerating arginine
E. Argininosuccinate synthetase, regenerating aspartate

40. A 29-year-old woman is diagnosed with celiac disease on serology and duodenal biopsy. She asks how her condition will be monitored once she starts treatment.

Which is the best way to monitor her response to a gluten-free diet?

A. Rising HLA-DQ2 expression
B. Repeat D-xylose test every month
C. Serial fecal elastase
D. Falling IgA anti-tTG levels
E. Annual colonoscopy

41. A 26-year-old woman has bloating, foul-smelling stools, iron-deficiency anemia, and an itchy vesicular rash on her elbows and knees. Duodenal biopsy shows villous atrophy, crypt hyperplasia, and intraepithelial lymphocytes.

This condition is most strongly associated with which HLA types?

A. HLA-DR5
B. HLA-B27
C. HLA-DQ2 and DQ8
D. HLA-DR3 and DR4
E. HLA-DR2

42. A 52-year-old woman has a positive anti-HCV antibody on routine one-time screening. She feels well, and her liver enzymes are normal.

Which is the most appropriate next step?

A. Start direct-acting antiviral therapy
B. HCV RNA by PCR
C. Repeat anti-HCV antibody in 6 months
D. Reassure her that she is immune
E. Liver biopsy

43. A 14-year-old boy with sickle cell disease has recurrent right upper quadrant pain after meals. An abdominal x-ray shows several radiopaque stones in the right upper quadrant.

Which best explains the formation of these stones?

A. Calcium oxalate precipitation
B. Bacterial deconjugation of bilirubin in infected bile ducts
C. Increased bile salt secretion
D. Cholesterol supersaturation of bile
E. Increased unconjugated bilirubin from chronic hemolysis

44. Two hours after eating leftover fried rice from a takeaway shop, a 25-year-old man develops nausea and repeated vomiting. His symptoms resolve within 24 hours.

Which organism is most likely responsible?

A. Staphylococcus aureus
B. Bacillus cereus
C. Salmonella enteritidis
D. Enterotoxigenic E. coli
E. Clostridium perfringens

45. A student lists hormones and hormone precursors made by the liver.

Which belongs on this list?

A. Aldosterone
B. Angiotensinogen
C. Insulin
D. Renin
E. Erythropoietin

46. A 38-year-old woman has dysphagia to solids and liquids and occasional regurgitation. Barium swallow suggests achalasia, and upper endoscopy shows no mass at the gastroesophageal junction.

Which test is the gold standard for confirming the diagnosis?

A. Esophageal manometry
B. Endoscopic ultrasound
C. CT chest
D. 24-hour esophageal pH monitoring
E. Repeat barium swallow

47. A 30-year-old pregnant woman with mild heartburn is started on famotidine.

Famotidine reduces acid secretion by which mechanism?

A. Activating prostaglandin E1 receptors
B. Irreversibly blocking the proton pump in parietal cells
C. Blocking muscarinic M3 receptors
D. Binding bile salts
E. Blocking H2 receptors, lowering cAMP in parietal cells

48. A 24-year-old man has right-sided abdominal pain and fever. Tenderness at McBurney point is mild, but passively extending his right hip while he lies on his left side reproduces the pain.

Where is the inflamed appendix most likely to lie?

A. In an inguinal hernia sac
B. In the pelvis, beside the obturator internus
C. Under the right lobe of the liver
D. Behind the cecum
E. Anterior to the terminal ileum

49. A 67-year-old woman with postprandial abdominal pain, fear of eating, and a 10-kg weight loss has severe atherosclerotic stenosis of the SMA and celiac trunk on CT angiography.

Which is the most appropriate treatment?

A. Long-term warfarin alone
B. Small frequent meals with nutritional support only
C. Endovascular angioplasty and stenting
D. Proton pump inhibitor
E. Cholecystectomy

50. A 72-year-old man with a lifelong low-fiber diet and chronic constipation is found to have many outpouchings in the sigmoid colon on colonoscopy.

Which statement best describes these outpouchings?

A. True diverticula containing all layers of the bowel wall, including the muscularis externa
B. Full-thickness herniation through the linea alba
C. Outpouchings of the posterior pharynx above the cricopharyngeus
D. False diverticula of mucosa and submucosa herniating at vasa recta entry points
E. Congenital remnants of the vitelline duct

Test C answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	**14** A B C D E	**27** A B C D E	**40** A B C D E
2 A B C D E	**15** A B C D E	**28** A B C D E	**41** A B C D E
3 A B C D E	**16** A B C D E	**29** A B C D E	**42** A B C D E
4 A B C D E	**17** A B C D E	**30** A B C D E	**43** A B C D E
5 A B C D E	**18** A B C D E	**31** A B C D E	**44** A B C D E
6 A B C D E	**19** A B C D E	**32** A B C D E	**45** A B C D E
7 A B C D E	**20** A B C D E	**33** A B C D E	**46** A B C D E
8 A B C D E	**21** A B C D E	**34** A B C D E	**47** A B C D E
9 A B C D E	**22** A B C D E	**35** A B C D E	**48** A B C D E
10 A B C D E	**23** A B C D E	**36** A B C D E	**49** A B C D E
11 A B C D E	**24** A B C D E	**37** A B C D E	**50** A B C D E
12 A B C D E	**25** A B C D E	**38** A B C D E	
13 A B C D E	**26** A B C D E	**39** A B C D E	

Test A answers and explanations

1 **D**	2 **B**	3 **B**	4 **D**	5 **A**	6 **A**	7 **C**	8 **A**	9 **B**	10 **C**
11 **D**	12 **A**	13 **E**	14 **E**	15 **A**	16 **B**	17 **A**	18 **B**	19 **E**	20 **B**
21 **E**	22 **D**	23 **C**	24 **B**	25 **C**	26 **D**	27 **A**	28 **B**	29 **D**	30 **B**
31 **E**	32 **E**	33 **B**	34 **E**	35 **A**	36 **D**	37 **C**	38 **C**	39 **D**	40 **D**
41 **E**	42 **C**	43 **A**	44 **C**	45 **C**	46 **C**	47 **A**	48 **E**	49 **D**	50 **E**

1. D. Urgent ERCP for biliary drainage

Ascending cholangitis · Step 2

Charcot triad (fever, RUQ pain, jaundice) plus hypotension and confusion is Reynolds pentad: severe ascending cholangitis with septic shock. ABX alone are not enough; the infected, obstructed bile must be drained urgently, usually by ERCP.

2. B. Anti-centromere antibody

Autoimmune liver injury · Step 1

PBC is associated with other autoimmune diseases, including limited systemic sclerosis (CREST: calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, telangiectasia), which is linked to anti-centromere antibodies.

D. Anti-Scl-70 is associated with diffuse systemic sclerosis.

3. B. Microvesicular steatosis

AFLP · Step 1

AFLP shows microvesicular steatosis: tiny fat droplets fill hepatocytes without displacing the nucleus, reflecting mitochondrial failure of fatty acid oxidation.

A. Macrovesicular steatosis with Mallory-Denk bodies suggests alcoholic liver disease.

E. Periportal fibrin deposits and hemorrhage suggest HELLP or pre-eclampsia.

4. D. Stopping cannabis

Physiology · Emetic center · Step 2

Years of cannabis use, cyclical vomiting relieved by hot showers, and wellness between episodes suggest cannabinoid hyperemesis syndrome. It resolves after stopping cannabis.

A. Cyclic vomiting syndrome is linked with migraine, but the hot showers and daily cannabis point elsewhere.

C. Scopolamine prevents motion sickness.

5. A. Skin, then liver, then kidney

Physiology · Liver function · Step 1

The skin makes cholecalciferol, the liver forms 25-OH cholecalciferol, and the kidney forms active calcitriol.

6. A. PPI + clarithromycin + metronidazole

PUD · Step 2

In penicillin allergy, metronidazole replaces amoxicillin in triple therapy, giving PPI + clarithromycin + metronidazole.

E. Amoxicillin is a penicillin and is contraindicated after anaphylaxis.

7. C. Interferon-gamma release assay (or tuberculin skin test)

IBD · Step 2

Infliximab is a TNF-α inhibitor. TNF-α is essential for maintaining granulomas, so blocking it can reactivate latent tuberculosis. Screen with an IGRA or tuberculin skin test (and for hepatitis B) before starting.

B. TPMT activity is checked before azathioprine or 6-MP, not before anti-TNF therapy.

8. A. It increases cholesterol secretion into bile

Cholelithiasis · Step 1

Estrogen increases hepatic cholesterol uptake and synthesis (upregulating HMG-CoA reductase), so more cholesterol enters bile. Progesterone in pregnancy also slows gallbladder emptying. Both help explain the "female" and "fertile" Fs.

9. B. Conjugated bilirubin reaching the urine

Physiology · Liver function · Step 2

Only conjugated bilirubin reaches the urine, so dark urine points to conjugated hyperbilirubinemia. Stool is pale if bile is obstructed.

10. C. Foveolar and mucous neck cells

Physiology · Gastric secretion · Step 1

Foveolar cells at the surface and in gastric pits, and mucous neck cells in the gland neck, secrete the mucus that forms the mucosal barrier.

11. D. Hepatitis E virus

Infectious liver injury · Step 1

HEV spreads by the fecal-oral route (contaminated water) and is usually self-limited, but in pregnancy, particularly the third trimester, it can cause fulminant hepatitis with high mortality. It is a naked +ssRNA hepevirus.

C. HAV is also fecal-oral, but it is not associated with fulminant hepatitis in pregnancy.

12. A. Shigella

Diarrhea · Step 1

Shigella has a very low infectious dose (as few as 10 organisms), so it spreads easily person to person. It invades colonic mucosa, causing dysentery, and is non-motile and does not produce H_2S.

E. Salmonella is motile, produces H_2S, and usually comes from poultry or eggs.

13. E. Inhibition through thoracolumbar sympathetic fibers

Physiology · Gastric secretion · Step 1

Sympathetic fibers from the thoracolumbar outflow inhibit stomach activity, opposing the stimulation from efferent CN X.

14. E. Corticosteroids

Autoimmune liver injury · Step 2

IgG4-related sclerosing cholangitis often occurs with autoimmune pancreatitis and can mimic PSC or cancer. Unlike PSC, it responds well to corticosteroids.

15. A. Pancreaticoduodenectomy (Whipple procedure)

Biliary malignancies · Step 2

A resectable tumor in the pancreatic head or periampullary region (the "double duct sign" shows both ducts blocked) is treated with pancreaticoduodenectomy (Whipple procedure), usually with chemotherapy. Stenting relieves jaundice but does not treat the cancer.

D. Distal pancreatectomy is used for tumors of the pancreatic body and tail.

16. B. Intramuscular vitamin B12

Gastritis · Step 2

Autoimmune gastritis destroys parietal cells, so intrinsic factor is lost and B12 cannot be absorbed normally. Parenteral (IM) B12 replacement bypasses the problem; high-dose oral B12 is an alternative.

A. Folic acid alone can correct the anemia while the neurologic damage from B12 deficiency progresses.

17. A. Aprepitant

Physiology · Emetic center · Step 2

Acute CINV within 24 hours is driven by 5-HT released from the gut wall onto 5-HT3 receptors, while delayed CINV on days 2–5 depends on substance P. NK1 antagonists such as aprepitant block substance P in the area postrema and NTS and prevent delayed CINV.

B. Scopolamine blocks vestibular input and prevents motion sickness.

C. Granisetron is another 5-HT3 antagonist, the class that already failed to prevent the delayed phase.

E. Hyoscine butylbromide is a peripheral antispasmodic.

18. B. It competitively blocks histamine, decreasing cAMP

Physiology · Gastric secretion · Step 2

H_2R antagonists competitively block histamine at H_2 receptors, which decreases cAMP and H^+/K^+ ATPase activity. They decrease HCl, especially basal and nocturnal secretion.

19. E. Stellate cells becoming myofibroblasts

Physiology · Liver function · Step 1

During tissue damage, stellate cells become myofibroblasts that secrete collagen I, leading to fibrosis.

20. B. IV ABX and laparoscopic cholecystectomy during this admission

Cholecystitis · Step 2

This is acute calculous cholecystitis: prolonged RUQ pain, fever, Murphy sign, leukocytosis, typical ultrasound findings, and normal LFTs. Treat with supportive care and ABX, followed by early laparoscopic cholecystectomy, ideally within 72 hours.

C. ERCP is for common bile duct stones; normal bilirubin and ALP make this unlikely.

E. Cholecystostomy is reserved for patients too unwell for surgery or with acalculous cholecystitis.

21. E. Hepatopancreatic ampulla (of Vater)

Choledocholithiasis · Step 1

Gallstone pancreatitis occurs when a stone blocks the ampulla, where the common bile duct and pancreatic duct meet, causing pancreatic enzyme activation. A cholestatic LFT pattern with high lipase and gallstones on ultrasound suggests a gallstone cause; it is the most common cause of acute pancreatitis.

B. A cystic duct stone causes biliary colic or cholecystitis with normal LFTs and lipase.

22. D. High doses reach μ receptors in the NTS behind the blood-brain barrier, which suppress vomiting

Physiology · Emetic center · Step 1

Opioids stimulate D2 and μ receptors in the CTZ, which provokes vomiting at low doses. At high doses, or with a lipophilic drug such as fentanyl, they cross the blood-brain barrier to μ receptors in the NTS, which suppress vomiting.

C. Opioids stimulate, not block, D2 receptors in the CTZ.

23. C. Diverticulosis

Diverticular diseases · Step 1

Diverticulosis is the most common cause of major lower GI bleeding in older adults. It is typically painless, sudden, and large in volume because a vasa recta artery ruptures. Most bleeds stop on their own; colonoscopy finds and treats the source.

A. An anal fissure causes painful, small-volume bleeding.

B. Ischemic colitis causes pain with bleeding.

24. B. Adenocarcinoma of the distal esophagus

GERD · Step 1

Barrett esophagus is intestinal metaplasia, in which nonciliated columnar epithelium with goblet cells replaces stratified squamous epithelium. It can progress through dysplasia to adenocarcinoma of the distal esophagus, so patients need endoscopic surveillance.

A. Squamous cell carcinoma is linked to smoking and alcohol and typically arises in the upper or middle esophagus.

25. C. Noncaseating granulomas with transmural inflammation

IBD · Step 1

Crohn disease shows skip lesions, transmural inflammation (causing fistulas and strictures), noncaseating granulomas, and Th1-mediated inflammation. It most often involves the terminal ileum and spares the rectum. Smoking worsens Crohn disease.

B. Caseating granulomas suggest intestinal tuberculosis.

D. Inflammation limited to the mucosa and submucosa, with crypt abscesses, is typical of ulcerative colitis.

26. D. Repeat endoscopy every 3–5 years

GERD · Step 2

Barrett esophagus without dysplasia is followed with endoscopic surveillance every 3–5 years to detect dysplasia. Dysplasia shortens the interval and may need endoscopic eradication therapy.

27. A. Inflammation spread from the visceral wall to the parietal peritoneum

Localizing abdominal pain · Step 1

Early appendicitis causes visceral pain, dull and periumbilical, from stretch of the appendix wall. When inflammation reaches the parietal peritoneum, somatoparietal pain becomes sharp and localizes to McBurney point.

C. A ureteric stone causes colicky flank pain radiating to the groin, with hematuria.

D. Referred pain is felt far from the organ, such as the shoulder, not at the organ itself.

E. Perforation causes sudden, diffuse pain and a rigid abdomen.

28. B. Graded-compression ultrasound, followed by MRI if inconclusive

Appendicitis · Step 2

In pregnancy, ultrasound is used first to avoid radiation, then MRI without gadolinium if ultrasound is inconclusive. The appendix may be displaced upward by the uterus, so pain can be atypical. Delaying treatment risks perforation, which threatens the fetus.

29. D. Gilbert syndrome

Physiology · Liver function · Step 2

Gilbert syndrome is benign, with mildly decreased UGT activity and mild unconjugated jaundice with fasting, stress, or illness.

30. B. Reassurance; no biopsy or treatment needed

Liver tumors · Step 2

Peripheral nodular enhancement with centripetal fill-in is typical of a hemangioma, the most common benign liver tumor. Asymptomatic hemangiomas need no treatment, and biopsy is avoided because of bleeding risk.

31. E. Antacids

Physiology · Gastric secretion · Step 2

Antacids neutralize HCl in the lumen, increasing gastric pH and giving fast, short-acting relief.

32. E. CT angiography of the abdomen

Intestinal ischemia · Step 2

Pain out of proportion to examination in a patient with an embolic source suggests acute mesenteric ischemia. CT angiography is the diagnostic test of choice. Treatment is urgent revascularization (endovascular or surgical) with anticoagulation, plus resection of necrotic bowel.

33. B. Lactase replacement

Physiology · Digestion and absorption · Step 2

Lactase deficiency leaves lactose undigested, causing osmotic diarrhea and gas from bacterial digestion. Treatment is lactase replacement or a lactose-free diet.

34. E. No treatment

Reading liver function tests · Step 2

An isolated unconjugated bilirubin after fasting or illness, with normal enzymes and FBC, suggests Gilbert syndrome, with UGT1A1 activity at 30–50%. It needs no treatment.

B. Normal enzymes and a typical trigger make biopsy unnecessary.

C. A stone gives a conjugated rise with raised ALP and GGT.

35. A. IV third-generation cephalosporin plus IV albumin

Acute liver failure and cirrhosis · Step 2

An ascitic neutrophil count of 250/μL or more diagnoses spontaneous bacterial peritonitis. Treat with a third-generation cephalosporin (e.g. ceftriaxone or cefotaxime). IV albumin reduces the risk of hepatorenal syndrome. Long-term ABX prophylaxis follows.

D. SBP has no surgical source; laparotomy is for secondary peritonitis such as a perforation.

36. D. Anticoagulation and stopping the oral contraceptive

Vascular liver injury · Step 2

All patients with Budd-Chiari syndrome are anticoagulated, and the underlying cause is treated (here, stopping estrogen). Treatment then steps up if needed: angioplasty or stenting for short venous narrowing, then TIPS, then liver transplant for fulminant failure or cirrhosis.

37. C. Serum triglycerides of 2400 mg/dL

Acute pancreatitis · Step 1

Hypertriglyceridemia (usually over 1000 mg/dL) is a cause of acute pancreatitis (the H in I GET SMASHED). Lipase breaks down the triglycerides into toxic free fatty acids that injure the pancreas. Milky (lipemic) serum is a clue. Treatment includes insulin infusion to lower triglycerides.

D. Hypercalcemia (not low calcium) causes pancreatitis; low calcium is a complication of it.

38. C. The nucleus tractus solitarius

Physiology · Emetic center · Step 1

The CTZ, vagal afferents, the vestibular system, and higher centers converge on the NTS, which begins the final common pathway and drives the central pattern generator in the medullary reticular formation.

39. D. Bile salts entering the colon directly

Biliary fistulas · Step 1

A cholecystocolic fistula lets bile bypass the small intestine, where bile salts are normally reabsorbed. Bile salts reaching the colon stimulate water secretion, causing bile acid diarrhea; bile also fails to reach the small bowel, so fat absorption suffers.

40. D. Hepatocyte copper excretion into bile

Metabolic liver injury · Step 1

Wilson disease is caused by ATP7B mutations (autosomal recessive). ATP7B moves copper into bile and onto ceruloplasmin. Without it, copper builds up in the liver and then deposits in the basal ganglia (movement and psychiatric symptoms) and cornea (Kayser-Fleischer rings).

41. E. Upper endoscopy

Esophageal motility disorders · Step 2

Older age, a short history, and marked weight loss suggest pseudoachalasia from a tumor at the gastroesophageal junction. Endoscopy must exclude malignancy before achalasia is diagnosed or treated.

42. C. IgA anti-tTG antibodies with total serum IgA

Malabsorption · Step 2

Screen for celiac disease with IgA anti-tTG while the patient is still eating gluten. Check total IgA because IgA deficiency (more common in celiac disease) gives false-negative results. Confirm with EGD and duodenal biopsy.

B. Biopsy confirms the diagnosis but follows a positive serology.

43. A. Therapeutic phlebotomy

Metabolic liver injury · Step 2

Regular phlebotomy is first-line: each unit of blood removes iron, and it is continued until ferritin is in the target range. Chelation is reserved for patients who cannot tolerate phlebotomy, such as those with anemia or heart failure.

D. D-penicillamine chelates copper and is used for Wilson disease.

44. C. Pregnant patients

PUD · Step 1

Misoprostol (a PGE1 analog) increases mucus and bicarbonate and reduces acid. It also causes uterine contractions and can induce abortion, so it is contraindicated in pregnancy. Its other main side effect is diarrhea.

45. C. Pseudohyphae and budding yeast

Esophagitis · Step 1

Candida esophagitis, the most common infectious esophagitis, shows pseudohyphae and budding yeast. Inhaled steroids, ABX, diabetes, and immunosuppression are risk factors.

A. Owl's eye inclusions suggest CMV.

D. Goblet cells in columnar epithelium describe Barrett esophagus.

E. Multinucleated giant cells suggest HSV.

46. C. CTR1, then ATP7A

Physiology · Digestion and absorption · Step 1

Copper enters the enterocyte through CTR1 and reaches the blood through ATP7A.

47. A. Decreased absorption in the duodenum

Physiology · Digestion and absorption · Step 1

Calcium is absorbed in the duodenum, and vitamin D increases absorption.

48. E. Replenishing glutathione to neutralize NAPQI

Iatrogenic and toxic liver injury · Step 1

In overdose, glucuronidation and sulfation are saturated, so more paracetamol is converted by CYP2E1 to the toxic metabolite NAPQI. Once glutathione is used up, NAPQI damages hepatocytes (zone 3 necrosis). N-acetylcysteine restores glutathione and is most effective within 8 hours.

C. Binding the drug in the gut describes activated charcoal, which helps only within 1–2 hours.

49. D. Dry mouth and constipation

IBS · Step 1

TCAs block muscarinic receptors, causing dry mouth, constipation, urinary retention, and blurred vision. They also cause sedation (via H1) and orthostasis (via α1). The constipating effect can be useful in IBS-D, and TCAs reduce visceral hypersensitivity.

50. E. CT abdomen with IV contrast

Traumatic liver injury · Step 2

A hemodynamically stable patient with suspected blunt abdominal trauma gets a contrast CT to identify and grade solid-organ injury. Most stable liver injuries are managed without surgery, with angioembolization if there is active bleeding.

D. Emergency laparotomy is for unstable patients with a positive FAST or peritonitis.

Test B answers and explanations

1 **A**	2 **B**	3 **C**	4 **D**	5 **B**	6 **E**	7 **C**	8 **C**	9 **D**	10 **B**
11 **B**	12 **B**	13 **A**	14 **C**	15 **B**	16 **C**	17 **C**	18 **D**	19 **A**	20 **D**
21 **E**	22 **E**	23 **E**	24 **A**	25 **A**	26 **E**	27 **D**	28 **E**	29 **C**	30 **A**
31 **C**	32 **B**	33 **D**	34 **C**	35 **E**	36 **D**	37 **B**	38 **E**	39 **D**	40 **E**
41 **E**	42 **A**	43 **B**	44 **A**	45 **C**	46 **B**	47 **D**	48 **D**	49 **A**	50 **A**

1. A. Zinc

Physiology · Digestion and absorption · Step 2

Zinc blocks intestinal copper absorption and is used for maintenance. D-penicillamine and trientine are copper chelators.

2. B. Unconjugated, as it is bound to albumin

Physiology · Liver function · Step 2

Unconjugated bilirubin is water insoluble and bound to albumin, so it does not reach the urine.

3. C. MRCP

Autoimmune liver injury · Step 2

Primary sclerosing cholangitis affects intra- and extrahepatic bile ducts, so it is diagnosed by cholangiography: MRCP shows multifocal strictures and dilations ("beading"). Liver biopsy is reserved for suspected small-duct PSC with a normal MRCP.

D. A HIDA scan evaluates acute cholecystitis.

4. D. Its capillaries lack tight junctions, so it sits outside the blood-brain barrier

Physiology · Emetic center · Step 1

The CTZ in the area postrema is a circumventricular organ whose capillaries lack tight junctions, so it can detect toxins in both blood and CSF. It lies on the floor of the fourth ventricle.

B. The area postrema lies on the floor of the fourth ventricle.

E. Vagal afferents carry signals from the gut to the NTS, not the other way.

5. B. Goal-directed IV fluids such as Hartmann solution

Acute pancreatitis · Step 2

Third spacing causes hypovolemia in pancreatitis. Early goal-directed IV fluids (Hartmann solution is commonly used), pain control and early feeding are the core of treatment. ABX are only for proven infection, and enteral feeding is preferred over TPN.

C. Prophylactic ABX do not improve outcomes; they are reserved for infected necrosis.

6. E. Increased estrogen from reduced hepatic clearance

Acute liver failure and cirrhosis · Step 1

The cirrhotic liver clears estrogen poorly, and more androgen is converted to estrogen peripherally. The excess estrogen causes gynecomastia, testicular atrophy, palmar erythema, and spider angiomas.

7. C. Bismuth subcitrate

Physiology · Gastric secretion · Step 2

Bismuth subcitrate coats the ulcer base, kills H. pylori, and increases PG and HCO_3^-.

8. C. Vibrio vulnificus

Metabolic liver injury · Step 1

Iron overload favors iron-loving (siderophilic) bacteria such as Vibrio vulnificus, Yersinia, and Listeria. Vibrio vulnificus from raw shellfish can cause fulminant sepsis with hemorrhagic bullae, so patients are told to avoid raw shellfish.

9. D. D2 blockade in the CTZ and 5-HT4 stimulation in the gut

Physiology · Emetic center · Step 1

Metoclopramide blocks D2 receptors in the CTZ and stimulates 5-HT4 receptors in the gut, releasing ACh, which speeds gastric emptying and improves upper GI motility.

A. Muscarinic blockade would slow, not speed, gastric emptying.

10. B. Oral rehydration, with loperamide if needed

Diarrhea · Step 2

Mild traveler's diarrhea, most often from enterotoxigenic E. coli, is self-limited. Oral rehydration is the mainstay, with loperamide for symptom control if there is no fever or blood. ABX are reserved for moderate to severe cases.

11. B. Elimination of food allergens from the diet

Esophagitis · Step 2

Eosinophilic esophagitis is an allergic, food antigen-driven inflammation of the esophagus. Elimination diets remove the trigger; PPIs and swallowed topical corticosteroids are also first-line options.

A. Pneumatic dilation of the LES treats achalasia.

12. B. Non-occlusive mesenteric ischemia

Intestinal ischemia · Step 1

Non-occlusive mesenteric ischemia follows low cardiac output and splanchnic vasoconstriction (shock, vasopressors, digoxin), without a blockage in the vessel. Treatment focuses on improving perfusion; intra-arterial vasodilators may help.

13. A. Cholangiocarcinoma

IBD · Step 1

Primary sclerosing cholangitis is strongly associated with ulcerative colitis (p-ANCA positive) and greatly raises the risk of cholangiocarcinoma. It also raises colorectal cancer risk further, so patients with both need closer colonoscopic surveillance.

14. C. Squamous cell carcinoma of the esophagus

Esophageal motility disorders · Step 1

Long-standing achalasia raises the risk of esophageal squamous cell carcinoma, probably from chronic irritation by stagnant food. Aspiration pneumonia is the other key complication.

15. B. ERCP with stone extraction

Choledocholithiasis · Step 2

Bilirubin over 4 mg/dL with a dilated CBD gives a high likelihood of choledocholithiasis, so proceed directly to ERCP to remove the stone, followed by cholecystectomy during the same admission. MRCP or endoscopic ultrasound is used when the likelihood is intermediate.

A. Cholecystectomy alone leaves the CBD stone in place.

16. C. Upper endoscopy with biopsy

GERD · Step 2

Dysphagia, weight loss, and iron-deficiency anemia are alarm features that require endoscopy with biopsy to exclude malignancy, rather than an empiric PPI trial.

17. C. Enterolithotomy to remove the stone from the bowel

Biliary fistulas · Step 2

Gallstone ileus is treated by enterolithotomy (opening the bowel to remove the stone), which relieves the obstruction. Cholecystectomy and fistula repair are usually deferred, especially in frail patients, because a combined procedure carries higher risk.

18. D. Fever, right upper quadrant pain, jaundice

Ascending cholangitis · Step 1

Charcot triad is fever, right upper quadrant pain, and jaundice. Adding hypotension and altered mental status gives Reynolds pentad, which signals septic shock.

A. Painless jaundice with a palpable gallbladder suggests malignancy (Courvoisier law).

B. RUQ pain, Murphy sign, and fever without jaundice suggest acute cholecystitis.

19. A. Empiric oral fluconazole

Esophagitis · Step 2

Candida is the most common cause of infectious esophagitis. In an immunocompromised patient with odynophagia and oral thrush, empiric systemic fluconazole is appropriate. Endoscopy is reserved for patients who do not improve within about 72 hours.

B. Acyclovir treats HSV esophagitis (small "punched-out" ulcers); oral thrush points to Candida.

C. Ganciclovir treats CMV esophagitis (large, shallow linear ulcers).

E. EGD is indicated only if empiric antifungal therapy fails.

20. D. Stomach acid stops salivary amylase

Physiology · Digestion and absorption · Step 1

Salivary amylase is secreted active and starts on starch in the mouth, and stomach acid then stops it.

21. E. It needs acid to polymerize into a paste

Physiology · Gastric secretion · Step 2

Sucralfate polymerizes at pH under 4 into a paste that binds the ulcer base, so it works poorly with acid-suppressing drugs.

22. E. Hepatic adenoma

Liver tumors · Step 1

Hepatic adenomas are linked to oral contraceptives and anabolic steroids, and larger ones (over 5 cm) can rupture and bleed. Management includes stopping estrogen and resecting large or bleeding adenomas.

B. Focal nodular hyperplasia has a central stellate scar and rarely bleeds.

23. E. Hypovolemia leading to gastric mucosal ischemia

Gastritis · Step 1

Curling ulcers follow severe burns. Massive fluid loss causes hypovolemia and gastric mucosal ischemia, which leads to erosions. Burn = Curling ("cooking"); brain = Cushing.

C. Increased vagal stimulation explains Cushing ulcers after brain injury or raised ICP.

24. A. Body and fundus

Gastritis · Step 1

Autoimmune gastritis targets parietal cells, which sit in the body and fundus. Their loss reduces HCl and intrinsic factor, leading to B12 deficiency (pernicious anemia, megaloblastic anemia, neuropathy). It causes fundal atrophy rather than erosions. H. pylori gastritis typically involves the antrum.

C. The antrum is the classic site of H. pylori gastritis and contains G cells, not parietal cells.

25. A. Biliverdin reductase

Physiology · Liver function · Step 1

Biliverdin reductase forms unconjugated bilirubin, which albumin carries to the liver.

26. E. Hypoglycemia with prolonged INR

AFLP · Step 2

AFLP causes true liver failure, so hypoglycemia and coagulopathy (raised INR, low fibrinogen) are common. HELLP is defined by hemolysis, raised liver enzymes, and low platelets, usually with hypertension, and rarely causes hypoglycemia. Both are treated with delivery.

27. D. Retrocecal, against the iliopsoas muscle

Appendicitis · Step 1

The psoas sign (pain on passive hip extension) means a retrocecal appendix is irritating the iliopsoas. The obturator sign (pain on internal rotation of the flexed hip) suggests a pelvic appendix. Pain migrating from the umbilicus to the RLQ reflects visceral pain becoming parietal as inflammation reaches the peritoneum.

C. A pelvic appendix gives a positive obturator sign.

28. E. Blood glucose

Acute liver failure and cirrhosis · Step 1

Hypoglycemia is one of the "big 4" complications of acute liver failure, alongside encephalopathy, coagulopathy, and infection. The failing liver cannot make glucose or release it from glycogen. A sudden drop in consciousness needs an immediate glucose check.

29. C. Phrenic nerve

Traumatic liver injury · Step 1

Blood or bile under the right hemidiaphragm irritates the diaphragmatic peritoneum, which is supplied by the phrenic nerve (C3–C5). Pain is referred to the C4 dermatome over the shoulder.

30. A. Emergency surgical repair

PUD · Step 2

Free air with peritonitis means a perforated ulcer, which is a surgical emergency. Resuscitate with IV fluids, IV PPI, and broad-spectrum ABX, then proceed to surgery.

B. Barium leaking into the peritoneum causes severe peritonitis; water-soluble contrast is used if contrast is needed at all.

C. Endoscopy insufflates air and can worsen a known perforation.

31. C. Serum AQP4 antibody

Physiology · Emetic center · Step 2

Area postrema syndrome, intractable nausea, vomiting, and hiccups, occurs when NMOSD attacks the AQP4-rich area postrema. With previous severe optic neuritis, AQP4 antibody testing is the key test.

B. Gastroparesis does not explain hiccups or the history of optic neuritis.

E. Cannabinoid hyperemesis causes cyclical vomiting relieved by hot showers.

32. B. No, colonoscopy is avoided during acute diverticulitis

Localizing abdominal pain · Step 2

LLQ pain, fever, and leukocytosis with these CT findings point to diverticulitis. Colonoscopy is avoided in peritonitis and acute diverticulitis, and is not needed for uncomplicated abdominal pain under 4 weeks.

C. CT AP with contrast is the first test for suspected diverticulitis in adults.

D. Screening starts at 45–50; there is no age at which she is too old to need it for symptoms.

E. CT does not exclude a colonic tumor or other mucosal disease.

33. D. Ileocecal valve

Cholecystitis · Step 1

Gallstone ileus: a large stone erodes through a cholecystoenteric fistula (usually cholecystoduodenal) and obstructs the narrowest part of the small bowel, the ileocecal valve. Rigler triad on imaging is pneumobilia, small bowel obstruction, and an ectopic gallstone.

B. A stone obstructing the pylorus or duodenum causes Bouveret syndrome (gastric outlet obstruction), not distal small bowel obstruction.

34. C. Elective laparoscopic cholecystectomy

Cholelithiasis · Step 2

Recurrent biliary colic (transient cystic duct obstruction, normal LFTs, no fever) is an indication for elective laparoscopic cholecystectomy. Asymptomatic gallstones need no treatment, but symptomatic ones do, because attacks recur and complications such as cholecystitis can follow.

A. Reassurance alone is for asymptomatic gallstones.

E. ERCP removes common bile duct stones; her normal LFTs argue against choledocholithiasis.

35. E. Enteropathy-associated T-cell lymphoma

Malabsorption · Step 1

Poorly controlled celiac disease raises the risk of enteropathy-associated T-cell lymphoma of the small bowel, as well as small bowel adenocarcinoma. A strict gluten-free diet lowers this risk.

36. D. Alcoholic hepatitis

Iatrogenic and toxic liver injury · Step 2

Alcoholic hepatitis typically shows an AST:ALT ratio above 2:1 with AST rarely above 300 U/L, raised GGT, and macrocytosis. ALT stays relatively low partly because it needs pyridoxal phosphate (vitamin B6), which is often deficient in heavy drinkers.

C. Viral hepatitis usually causes ALT-predominant rises, often well above 1000 U/L.

E. MASLD typically has ALT higher than AST.

37. B. Reassurance and a trial of a low-FODMAP diet

IBS · Step 2

IBS is a disorder of gut-brain interaction diagnosed clinically (Rome V criteria) once alarm features are absent and basic tests are normal. First-line management is reassurance, education, and a low-FODMAP diet, with symptom-directed drugs as needed.

E. Colonoscopy is not needed without alarm features in a patient under 45.

38. E. Sucrase, forming glucose and fructose

Physiology · Digestion and absorption · Step 1

Sucrase converts sucrose to glucose and fructose, lactase converts lactose to glucose and galactose, and maltase converts maltose to glucose.

39. D. Hepatitis B vaccine plus HBIG within 12 hours of birth

Infectious liver injury · Step 2

Perinatal HBV transmission is highly likely to cause chronic infection (about 90%). Giving both the first vaccine dose and HBIG within 12 hours of birth prevents most infections.

40. E. Decreased micelles to carry vitamin K

Physiology · Digestion and absorption · Step 1

Vitamins A, D, E, and K travel in micelles and leave in chylomicrons, so bile salt deficiency causes fat-soluble vitamin deficiencies.

41. E. It displaces toxic bile salts and increases bile flow

Physiology · Liver function · Step 2

Ursodeoxycholic acid is a hydrophilic bile salt that displaces toxic hydrophobic bile salts and increases bile flow, slowing progression in PBC.

42. A. It replenishes glutathione

Physiology · Liver function · Step 2

GST joins GSH to toxins such as NAPQI. NAC replenishes glutathione, which is why it treats paracetamol overdose.

43. B. Blood reaches it last, so it has the least oxygen

Physiology · Liver function · Step 1

Blood reaches zone 3 last, so it receives the least oxygen and is first to suffer from low perfusion.

44. A. Raised central venous pressure transmitted to the hepatic veins

Vascular liver injury · Step 1

Right heart failure raises central venous pressure, which backs up into the hepatic veins and sinusoids. Zone 3 congestion and atrophy produce the mottled "nutmeg liver". Chronic congestion can lead to cardiac cirrhosis.

B. Portal vein thrombosis causes portal hypertension without centrilobular congestion.

D. Budd-Chiari syndrome also causes a nutmeg pattern, but JVD and hepatojugular reflux point to a cardiac cause.

45. C. ABX therapy

Diarrhea · Step 2

This is hemolytic uremic syndrome from Shiga toxin-producing E. coli (O157:H7). ABX, and antimotility agents such as loperamide, may increase toxin release and the risk of HUS, so they are avoided. Care is supportive: fluids, transfusion, and dialysis as needed.

46. B. Bone

Reading liver function tests · Step 1

ALP also comes from bone, placenta, and intestine. A raised ALP with a normal GGT points to bone, pregnancy, or growth; a high GGT would confirm a hepatobiliary source.

A. Obstruction raises GGT and usually bilirubin along with ALP.

E. PBC is cholestatic, so GGT would rise with ALP.

47. D. I cells secreting CCK

Physiology · Gastric secretion · Step 1

Fat and amino acids release CCK from I cells, which decreases HCl, slows gastric emptying, and causes gallbladder contraction and pancreatic enzyme release.

48. D. Splenic flexure

Intestinal ischemia · Step 1

Ischemic colitis from hypoperfusion affects watershed areas. The splenic flexure (between the SMA and IMA territories) and the rectosigmoid junction are most vulnerable. The distal rectum is spared because it has a dual blood supply from the IMA and the iliac branches.

E. The distal rectum has a dual blood supply and is usually spared.

49. A. CT-guided percutaneous drainage plus ABX

Diverticular diseases · Step 2

Diverticular abscesses larger than about 3–4 cm are drained percutaneously alongside ABX. Surgery is reserved for failed drainage, perforation with peritonitis, or recurrent complicated disease.

50. A. Clonorchis sinensis

Biliary malignancies · Step 1

A perihilar mass at the hepatic duct confluence (Klatskin tumor) is a cholangiocarcinoma. Liver flukes such as Clonorchis sinensis, acquired from raw freshwater fish, cause chronic bile duct inflammation and raise the risk. PSC and choledochal cysts are other risk factors.

E. Schistosoma haematobium is linked to squamous cell carcinoma of the bladder.

Test C answers and explanations

1 **A**	2 **A**	3 **B**	4 **A**	5 **B**	6 **D**	7 **B**	8 **E**	9 **E**	10 **D**
11 **A**	12 **C**	13 **A**	14 **E**	15 **A**	16 **E**	17 **B**	18 **B**	19 **D**	20 **C**
21 **E**	22 **C**	23 **B**	24 **C**	25 **A**	26 **B**	27 **A**	28 **D**	29 **A**	30 **C**
31 **D**	32 **D**	33 **C**	34 **E**	35 **D**	36 **C**	37 **E**	38 **E**	39 **C**	40 **D**
41 **C**	42 **B**	43 **E**	44 **B**	45 **B**	46 **A**	47 **E**	48 **D**	49 **C**	50 **D**

1. A. Colovesical fistula

Diverticular diseases · Step 2

Pneumaturia, fecaluria, and recurrent polymicrobial UTIs suggest a colovesical (enterovesical) fistula, most often from diverticulitis. CT is the usual first test (air in the bladder), and treatment is surgical.

2. A. It is an HDV superinfection and is likely to become chronic

Infectious liver injury · Step 1

HDV needs HBsAg to form its envelope, so it only infects people with HBV. A negative IgM anti-HBc shows the HBV is chronic, so this is a superinfection, which is usually more severe and becomes chronic. In coinfection, both are acquired together and IgM anti-HBc is positive.

B. Coinfection would show a positive IgM anti-HBc, because the HBV infection would be acute.

C. There is no hepatitis C vaccine; HDV is prevented by the hepatitis B vaccine.

3. B. The vestibular system

Physiology · Emetic center · Step 1

Motion sickness comes from mismatch between what the vestibular, visual, and proprioceptive systems report. People with total loss of labyrinthine function are immune. Lesioning the area postrema, by contrast, does not stop motion-induced vomiting.

E. The CTZ is not needed for vomiting from motion.

4. A. Azathioprine

Acute pancreatitis · Step 1

Drugs are the "D" in I GET SMASHED. Azathioprine is a classic cause of acute pancreatitis, usually within the first weeks of therapy. Others include valproate, thiazides, furosemide, sulfonamides, and didanosine.

5. B. It is a PG analog that decreases HCl and increases mucus

Physiology · Gastric secretion · Step 2

Misoprostol is a PG analog acting on EP receptors on parietal cells. It decreases HCl and increases mucus and HCO_3^-, giving gastroprotection in NSAID users.

6. D. No further test is needed

Liver tumors · Step 2

In a patient with cirrhosis, a lesion with arterial-phase enhancement and washout on multiphase CT or MRI is diagnostic of hepatocellular carcinoma; biopsy is not required. Biopsy is reserved for lesions with atypical imaging.

7. B. DMT1, ferroportin, then transferrin

Physiology · Digestion and absorption · Step 1

Iron enters the enterocyte through DMT1, reaches the blood through ferroportin, and is carried by transferrin.

8. E. Pancreatic enzyme replacement

Physiology · Digestion and absorption · Step 2

Pancreatic insufficiency decreases pancreatic enzymes, leaving fat, protein, and carbohydrate undigested. Enzyme replacement treats it.

9. E. Immune reaction to drug-modified liver proteins

Iatrogenic and toxic liver injury · Step 1

Halothane hepatitis is an immune-mediated reaction: trifluoroacetylated liver proteins act as neoantigens, so severe injury usually follows repeat exposure. Fever and eosinophilia are clues to an immune drug reaction.

10. D. Emergency laparotomy

Traumatic liver injury · Step 2

A hemodynamically unstable trauma patient with a positive FAST needs emergency laparotomy to control bleeding. Unstable patients should not be sent to CT.

11. A. Crigler-Najjar syndrome type I

Physiology · Liver function · Step 2

Crigler-Najjar syndrome type I has absent UGT activity, with severe neonatal jaundice and kernicterus. Type II is milder and responds to phenobarbital.

12. C. Gastric phase, about 60%

Physiology · Gastric secretion · Step 1

The cephalic phase contributes about 30% of HCl, the gastric phase about 60%, and the intestinal phase about 10%.

13. A. Cholecystectomy

Ascending cholangitis · Step 2

The gallbladder is the source of most CBD stones, so cholecystectomy (once she has recovered, usually during the same admission) prevents recurrent choledocholithiasis and cholangitis.

14. E. Protons and IF

Physiology · Gastric secretion · Step 1

Parietal cells in the body and fundus secrete protons and IF. In autoimmune gastritis, their loss decreases HCl and IF.

A. These are made by chief cells.

B. Mucus comes from foveolar and mucous neck cells.

C. Histamine comes from ECL cells.

D. These come from G cells and D cells.

15. A. Proteases freeing B12 in the duodenum

Physiology · Digestion and absorption · Step 1

B12 binds haptocorrin from saliva in the stomach. Proteases free it in the duodenum, where it binds IF from parietal cells.

16. E. Loperamide

IBD · Step 2

This is toxic megacolon. Antimotility agents (loperamide) and opioids worsen colonic dilation and can precipitate perforation. Management is bowel rest, IV fluids, IV steroids, ABX, and early surgical consultation. VTE prophylaxis is still needed because IBD is hypercoagulable.

17. B. Blocking acetylcholine release from nerve terminals

Esophageal motility disorders · Step 1

Botulinum toxin cleaves SNARE proteins and blocks acetylcholine release from excitatory neurons, lowering LES tone. The effect wears off over months, so it is used in patients who are poor candidates for dilation or myotomy.

C. Blocking L-type calcium channels is how calcium channel blockers such as nifedipine work.

18. B. Fibromyalgia

IBS · Step 1

IBS is a disorder of gut-brain interaction driven by visceral hypersensitivity, altered microbiome, and abnormal motility. It commonly coexists with fibromyalgia, anxiety, and depression.

19. D. Cholestyramine

Autoimmune liver injury · Step 1

Cholestyramine is a bile acid sequestrant. It interrupts enterohepatic circulation, lowering bile salts that build up in cholestasis and cause pruritus. Give it apart from other drugs (including UDCA), because it binds them too.

C. Ezetimibe blocks cholesterol absorption (NPC1L1) and is not used for pruritus.

20. C. Portal vein

Vascular liver injury · Step 1

The liver has a dual blood supply: the portal vein provides about 75% of hepatic blood flow, and the hepatic artery about 25% (but roughly half the oxygen). This dual supply makes hepatic infarction uncommon.

21. E. Long-chain 3-hydroxyacyl-CoA dehydrogenase (LCHAD)

AFLP · Step 1

AFLP is linked to fetal LCHAD deficiency, a defect in mitochondrial beta-oxidation of long-chain fatty acids. Fatty acid metabolites from the fetus overwhelm the heterozygous mother's liver, causing microvesicular steatosis.

22. C. Hypochloremic metabolic alkalosis

Physiology · Emetic center · Step 1

Repeated vomiting loses HCl, leaving its HCO_3^- in the blood and causing hypochloremic metabolic alkalosis. Her picture fits hyperemesis gravidarum, which starts at 4–6 weeks; thiamine prevents Wernicke encephalopathy.

D. Hyperchloremic acidosis follows loss of bicarbonate, as in diarrhea.

23. B. Echocardiography

Vascular liver injury · Step 2

Hepatomegaly with jugular venous distention and a positive hepatojugular reflux points to congestive hepatopathy from right heart failure, so echocardiography is the key test. Budd-Chiari syndrome causes hepatomegaly and ascites without JVD.

24. C. Right lower quadrant ultrasound

Appendicitis · Step 2

In children and pregnant patients, ultrasound is the first imaging test for suspected appendicitis to avoid radiation. CT with contrast is the most accurate test in adults, or if ultrasound is inconclusive.

25. A. Pancreatic head or other periampullary malignancy

Biliary malignancies · Step 2

Courvoisier law: painless jaundice with a palpable, nontender gallbladder is unlikely to be caused by gallstones. A slowly growing tumor below the cystic duct lets a healthy gallbladder distend. Gallstones usually scar the gallbladder, so it cannot enlarge, and they cause pain.

B. CBD stones usually cause painful jaundice, and the gallbladder is typically shrunken from chronic inflammation.

26. B. Transdermal scopolamine, blocking muscarinic input to the vestibular nuclei

Physiology · Emetic center · Step 2

Scopolamine, given as a transdermal patch, blocks muscarinic receptors and inhibits input to the vestibular nuclei, preventing motion sickness. H1 antagonists that act in the brain also help.

C. Second-generation antihistamines do not prevent motion sickness, likely because they act peripherally.

E. 5-HT3 antagonists target chemotherapy, radiation, and postoperative vomiting, not vestibular input.

27. A. Gastric adenocarcinoma

PUD · Step 1

CagA-positive H. pylori strains cause more severe inflammation and carry a higher risk of peptic ulcers and gastric adenocarcinoma. VacA is another cytotoxin associated with more severe disease.

28. D. Cholecystoduodenal

Biliary fistulas · Step 1

Air in the biliary tree without prior instrumentation suggests a biliary-enteric fistula. Cholecystoduodenal fistulas are the most common, because the gallbladder lies right next to the duodenum, and they are the usual route for gallstone ileus.

29. A. Transferrin saturation and serum ferritin

Metabolic liver injury · Step 2

Transferrin saturation (above 45% is abnormal) with ferritin is the initial screen for iron overload. HFE genetic testing follows, which is also offered directly to first-degree relatives. Liver MRI or biopsy is reserved for unclear cases.

30. C. Reabsorption into the portal vein

Physiology · Digestion and absorption · Step 1

Micelles work in the jejunum, then bile salts are reabsorbed in the terminal ileum and return to the liver through the portal vein.

31. D. G, ECL, parietal, and chief cells

Physiology · Gastric secretion · Step 1

Somatostatin from D cells inhibits G, ECL, parietal, and chief cells. It acts through SSTR2, Gi, and decreased cAMP.

32. D. Eosinophilic esophagitis

Esophagitis · Step 1

Eosinophilic esophagitis occurs in atopic patients (asthma, allergic rhinitis, atopic dermatitis) and is driven by food allergens. Classic findings are food impaction, "trachealized" rings and furrows, and at least 15 eosinophils/hpf. Management includes eliminating food allergens, PPIs, and swallowed topical steroids.

B. A Schatzki ring is a single distal mucosal ring and does not explain dense eosinophilic infiltration.

E. GERD can cause some eosinophils, but rings, furrows, atopy, and food impaction point to EoE.

33. C. Sodium restriction and spironolactone, with furosemide added as needed

Acute liver failure and cirrhosis · Step 2

First-line ascites treatment is dietary sodium restriction plus spironolactone (blocking secondary hyperaldosteronism), often with furosemide. Large-volume paracentesis with albumin, TIPS, or transplant are used for tense or refractory ascites.

34. E. GLUT5 into the cell, then GLUT2 into the portal vein

Physiology · Digestion and absorption · Step 1

Fructose enters through GLUT5. Glucose, galactose, and fructose all leave through GLUT2 into the portal vein.

35. D. A failing liver with fewer hepatocytes left to injure

Reading liver function tests · Step 2

Falling AST and ALT with rising bilirubin and INR mean a failing liver, not recovery. With encephalopathy and a high INR within 26 weeks and no prior liver disease, this is acute liver failure, which needs transplant referral.

A. An INR from vitamin K deficiency corrects with parenteral vitamin K and does not cause drowsiness.

B. Falling enzymes are reassuring only if bilirubin and INR are also improving.

C. Obstruction gives a cholestatic pattern with raised ALP and GGT.

36. C. IV ABX and percutaneous cholecystostomy

Cholecystitis · Step 2

Acalculous cholecystitis occurs in critically ill patients from gallbladder stasis and ischemia, and carries a high risk of gangrene and perforation. Unstable patients are treated with ABX and percutaneous drainage (cholecystostomy).

37. E. Bacterial deconjugation of bilirubin within the bile duct

Choledocholithiasis · Step 1

Brown pigment stones form within the bile ducts (primary CBD stones), where infection with bacteria producing β-glucuronidase deconjugates bilirubin, which precipitates with calcium. They can form years after cholecystectomy.

38. E. Prophylactic PPI or H2 blocker

Gastritis · Step 2

Critically ill patients, especially those on mechanical ventilation, with coagulopathy, burns, or head injury (Cushing ulcers), are at risk of stress ulcer bleeding. Acid suppression with a PPI or H2 blocker is used as prophylaxis.

39. C. Arginase, regenerating ornithine

Physiology · Liver function · Step 1

Arginase converts arginine to urea, releasing ornithine to restart the cycle.

40. D. Falling IgA anti-tTG levels

Malabsorption · Step 2

Treatment is a lifelong gluten-free diet (avoiding wheat, rye, and barley). Celiac antibodies such as IgA anti-tTG fall with good adherence, and persistently raised levels suggest ongoing gluten exposure.

A. HLA type does not change; HLA-DQ2/DQ8 testing helps rule celiac disease out, not monitor it.

41. C. HLA-DQ2 and DQ8

Malabsorption · Step 1

Celiac disease is associated with HLA-DQ2 and DQ8. Gliadin (from wheat, rye, and barley) is deamidated by tissue transglutaminase and presented on these HLA molecules, driving a T-cell response that damages the mucosa. The rash is dermatitis herpetiformis. Iron deficiency reflects duodenal involvement.

B. HLA-B27 is linked to the seronegative spondyloarthropathies.

D. HLA-DR3/DR4 is classically linked to type 1 diabetes.

42. B. HCV RNA by PCR

Infectious liver injury · Step 2

A positive anti-HCV antibody shows exposure but not whether the infection is current; the antibody stays positive for life, even after cure. HCV RNA confirms active infection. If positive, assess fibrosis (e.g. APRI, FIB-4, or elastography) before treatment.

D. Anti-HCV antibodies are not protective; reinfection is possible.

43. E. Increased unconjugated bilirubin from chronic hemolysis

Cholelithiasis · Step 1

Chronic hemolysis (sickle cell disease, hereditary spherocytosis) increases bilirubin production. The extra bilirubin precipitates with calcium in the gallbladder as black pigment stones, which are often radiopaque.

B. Bacterial deconjugation produces brown pigment stones, which form in the bile ducts.

44. B. Bacillus cereus

Diarrhea · Step 1

B. cereus spores survive cooking. In rice left at room temperature they germinate and form a preformed, heat-stable emetic toxin (cereulide), causing vomiting within 1–6 hours. This is called "reheated rice syndrome".

A. S. aureus also causes rapid vomiting from a preformed toxin, but the classic sources are mayonnaise, potato salad, and custards.

45. B. Angiotensinogen

Physiology · Liver function · Step 1

The liver makes TPO, IGF-1, angiotensinogen, and 25-OH cholecalciferol.

46. A. Esophageal manometry

Esophageal motility disorders · Step 2

Esophageal manometry is the gold standard. In achalasia it shows incomplete LES relaxation, raised LES pressure, and absent distal peristalsis. Endoscopy is done first to exclude malignancy at the GE junction (pseudoachalasia).

47. E. Blocking H2 receptors, lowering cAMP in parietal cells

GERD · Step 1

H2-receptor antagonists competitively block histamine H2 receptors on parietal cells, reducing cAMP and acid secretion. They are less potent than PPIs and suit milder symptoms.

48. D. Behind the cecum

Localizing abdominal pain · Step 1

The psoas sign, pain on passive extension of the right hip, suggests a retrocecal appendix lying on the psoas muscle. The obturator sign, pain on internally rotating the flexed hip, suggests a pelvic appendix.

B. A pelvic appendix gives the obturator sign.

49. C. Endovascular angioplasty and stenting

Intestinal ischemia · Step 2

Symptomatic chronic mesenteric ischemia is treated with revascularization, usually endovascular angioplasty and stenting (or open bypass), plus risk factor control. It relieves pain and prevents progression to acute bowel infarction.

50. D. False diverticula of mucosa and submucosa herniating at vasa recta entry points

Diverticular diseases · Step 1

Colonic diverticula are false (pulsion) diverticula. Mucosa and submucosa herniate through weak points in the muscularis externa where the vasa recta enter. High intraluminal pressure in the sigmoid, from low fiber, constipation, and aging, drives their formation.

C. An outpouching above the cricopharyngeus is a Zenker diverticulum.

E. A vitelline duct remnant is a Meckel diverticulum, which is a true diverticulum.